

Universe24: one integer rule at every Node of a lattice: the click as the local, whole measurement, the cloud as the body of nature, and the forms of Einstein and Lorentz arrived at without them as inputs

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Abstract

We state one local rule in whole numbers, Rule3, applied at every Node of a cubic lattice each interval: a second-order step of a family's level from its two levels and its six neighbours' arrivals, with integer coefficients from the family's pair and the Node's paces, and one Euclidean division whose remainder stays at the Node. Every level on the lattice is written by that one act. A quantum moves whole when a remainder crosses the quantum's action, and that division is the click, the one measurement, local and whole: the record goes on after it, nothing ends anywhere, and no act at a distance is in the law. Exact on the lattice: the whole run returns bit for bit under the rule's own direction, a form is conserved, the count is conserved, and the momentum's flux is the wave's own stress. The families are bands of the rule. The body of nature is the cloud, the wide fixed point of the binding row, whose three masses (the well it sources, the count it resists with and the tail it falls with) are one to the order $(C/\Gamma)^2$; the compact pixel is collapsed matter. From the same coefficients, with a held level lowering the clock's pace once and the Link's pace once more, come the gravitational redshift, light's bending by twice the clock's share, the delay, the perihelion's 6π once the clock carries its second order, and Lorentz's factor at a body's own light speed; each form stands on the comparison side and none is an input. The first experiment run on the engine, the two slits, gives the fringes at the formula's spacing from the local click alone, beside a blind row written before the run, and two findings written as they stand: a dilute packet on a cold board, every remainder 0, reports nothing, and a bare detector in one gap changes nothing. On a warm board a detector's sensitivity is its remainder and Born's rule for the detectors follows from the count's line; the price of locality is stated plainly: two far detectors both rise on one quantum with the probability $f_1 f_2$, which nature never shows. The paper claims that the detectors' clicks represent reality and shows the match; it never claims that nature is so.

Keywords: integer lattice model; one local rule; the click as the measurement; locality; discrete relativity; pre-registered simulation

1 Introduction

The question. The law of this paper is one line of integer arithmetic applied at every Node of a cubic lattice at every interval. Nothing else acts: no field equation beside it, no law of motion for a body, no rate, no draw, no real number. The paper states that line, proves what is exact on the lattice, derives what follows from the line alone, names what is decided beside it, and sets the law's own numbers beside the forms of nature: the forms of Einstein, Lorentz, Newton, de Broglie and

Young appear only on the comparison side, beside the law's expectation written before any run; none is an input. Every formula is taken from the law's document [1] and cited by the name of the section it stands in; the engine that computes the rule is described by its own document [2], and the decisions in force by theirs [3].

The click's story, the spine of the paper. A click is the passage of one whole quantum from one Node to the next, at one Link, where a remainder kept at the Node crosses the quantum's action: the count's line's own division (Section 3.2). It is local, it is whole, and it ends nothing. The record, the wave on the lattice, goes on after it exactly as before; no wave collapses, no record ends anywhere, and no act at a distance is in the law. A detector is a Node declared in a file to report the rise of a family's count across its Nodes; it takes nothing and hands nothing over. Nature's detector, the paper says, is a body: a body reads the light's level at its Nodes by its own rotation sense into its paces, a drive at the light's frequency, resonant where it meets the body's rotation, and its write answers, the driven body writing light of its own that cancels the incoming wave behind it; where the wave is cancelled its current stops, the light's count that arrived at the body's Nodes does not leave, the count's line piles it there, and a Node told to report reports the rise: one photon at one resonant absorber clicks once, locally (Section 3.6). That is the whole story of measurement in this law, and its price is stated plainly in Section 3.7: a detector's sensitivity is its remainder, so on a warm board the chance of a rise is the intensity's share, Born's rule from the count's line, but two far detectors do not know of one another and both rise on one quantum with the probability $f_1 f_2$, which nature never shows; a cold detector Node, every remainder 0, reports no spread quantum; and Bell's row is open. The paper calls these findings against nature and writes them as they stand.

The reading rule. Inside is inside the GameBoard, where no one measures; Outside is the game above the board, the detectors and their clicks. Only a detector's click is a measurement, compared with nature or pinned as an expectation. A number of the GameBoard's state (a level, a support, a total, a body's centre) is a GameBoard reading, a diagnostic, labelled so wherever it appears, never compared with nature and never a result on its own; a number whose kind is not named is not a result [1, readings and measurements]. The paper claims that the Outside represents reality and shows the match; it never claims that nature is so [3, inside and outside].

The marks. Every claim carries one of five marks: THEOREM, proved for every GameBoard; DERIVED, from theorems by argument; COMPUTED, a number of the law's own computed from its lines before any run; ASSUMPTION, a line the law holds by decision and names where it is used; HYPOTHESIS, a line outside the law under its own name, with what would decide it. A number of a run enters only through the documents' rows, labelled DETECTOR (a click, the measurement) or GAMEBOARD (a diagnostic).

The law in one line. At every Node, each interval, for every family [1, the line; the count's line; the primitives]:

Rule3:	$w a_{\text{next}} + r' = \sum_{a \in \{x, y, z\}} R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{ den } \Gamma^2, \quad R_a = 2 \text{ num } p_a^2,$ $S = 12 \text{ den } \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - 4 \text{ num } (p_x^2 + p_y^2 + p_z^2);$	(1)
the count's line:	$W_c c_{\text{next}} + r' = W_c c_{\text{now}} + \sum_j F_{ij} + r, \quad 0 \leq r' < W_c, \quad W_c = 3 \text{ den } T,$ $F_{ij} = \text{num} (\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j);$	
the one write:	a level at a Node gains $(w q(\text{Node}) + r) \text{ div } E$, the remainder r kept there.	

In the first line a_{now} , a_{before} and a_{next} are the family's level at the Node now, one interval before and one interval on; $\text{arr}_{\pm a}$ is the arrival through the Port $\pm a$, the neighbour's level; r and r' are

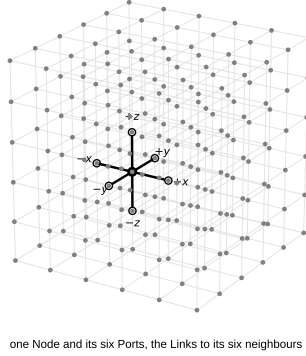


Fig. 1 The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x, \pm y, \pm z$ (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it, whose convex hull is the octahedron of Fig. 2. A Node keeps each family's two levels and its remainders there, and nothing else. Drawn from the definitions; no run.

the remainder kept at the Node before and after the step; Γ is the Node clock, the pace of an empty Node; $[\text{num}, \text{den}]$ is the family's pair, with $\cos \omega_0 = \text{num}/\text{den}$ its rotation per interval at wave number zero; p_0 is the Node's clock pace and p_a its pace along the axis a , both the Node clock lowered by the levels the family reads there; and div is the floor division. The second line is the same rule for the family's quanta: c is the count, the number of quanta at the Node; T is the quantum's action, the one unit of every family; F_{ij} is the current into the Node i from its neighbour j over the interval, the bilinear form of the record's two levels at the two ends; and the division is the click, one quantum moving whole exactly when the remainder crosses the wall W_c . The third line is every write of a record into a held level: $q(\text{Node})$ is the writer's own quantity there, w the one weight with which the writer's family reads the written family, and E the written row's divisor. Everything else is a number in a file. The first line is a theorem's object (Section 2), the second the click (Section 3), the third the one write (Section 2.8).

The words. The physical location is a *Node*; its complete local information is its *NodeState*, the law's own numbers and nothing else. Nodes are joined by *Links* through six directional *Ports* per Node; the lattice of Nodes is the *GameBoard*; a local change is an *Event*, and there is one *LocalRule*, Rule3. A *Family* is one kind of physical value with its pair and what it holds; its *Record* is its levels over the GameBoard, two per Node with a remainder. A *Body* is the Nodes on which a family's count stands with the standing record the count binds, with no law of its own; a *Message* is a record the world lays and no body. A *Detector* is a Node declared in the file to report; its *Click*, the rise of a family's count across its Nodes, is the one measurement. An *Interval* is one step of the whole GameBoard [1, the objects][2, the words].

What this paper claims, and what it does not. Claimed: one explicitly stated integer rule whose reversibility, conserved form, count conservation and momentum's flux are theorems on the lattice; the click as the local, whole measurement, with the record never ending; the families as bands of that rule; the cloud as the body of nature with its three masses one, and the compact pixel as collapsed matter; the forms of nature arrived at from the coefficients and set beside their known forms; and one experiment's row with its blind expectation and its result. Not claimed: that nature is so; a derivation of the Lorentz group, which is no symmetry of the lattice (the lattice's symmetries are the 48 signed permutations of the axes); Bell's correlation, which the local click

does not give and which stands as an open row, and the anticorrelation of two takers, which the law gives the other way; the fine-structure constant, the mass ratio of the proton to the electron or what fixes the Node clock, which are open by name (Section 8.4). The paper holds no run of its own; every number of a run is the documents' and is cited so.

2 The general formula

This section states the object and proves what is exact on it: the lattice and its families, the one line with its coefficients, its reversibility, what its acts can and cannot form, the paces with their guard, the conserved form, the tension as the wave's own stress, the one write, the interval and its inverse, and the three channels through which one family acts on another [1, the GameBoard and the families; Rule3; the primitives].

2.1 The objects

A Node is a physical location. The GameBoard is all the Nodes together, a box of (N_x, N_y, N_z) Nodes; each axis is periodic (the box a torus on that axis), open (a face beyond which no Node lies, read as 0) or folded (an axis of one layer, whose two Ports return the Node itself), as the world declares. Two Nodes that differ by one step along one axis are joined by a Link; each Node has six Links, one through each of its six Ports, $\pm x, \pm y, \pm z$ (Fig. 2). An inner face is the face's rule declared inside the board: a plane of Nodes across an axis, beyond the board but for the gaps the world declares; a Node beyond the board is read as 0 through every Port and reads 0 through its own (a held row with a declared rest, the massless row at the file's c_{vac} , is read at its rest there, the vacuum beyond the face being the same vacuum), so no level, current or source crosses its Links, and no body, message or detector stands there. The board's shape is a declaration of the file, like the wraps and the detector Nodes, and no number of the engine. The local information of a Node is its NodeState: the levels and remainders of the families at that Node, and nothing else; no register, no table, no formula and no draw stand at a Node [1, the objects][3, nothing at a Node but the law's numbers].

A family is one kind of physical value with its own declarations: its pair [num, den] and what it holds. Its record is two levels per Node (now and before) with a remainder per Node; a family of quanta carries two such level pairs, the real one and its second, the rotation sense (Section 3.4), and a count with the count's remainder. A body is a set of Nodes on which a family's count stands, one connected region; a message is a record the world lays and no body, a packet of a family of quanta with its axis, its wave number, its amplitude and its envelope declared [1, the objects].

2.2 The line

At every Node, each interval, every family steps by the first line of Eq. (1): the six arrivals, each the neighbour's level through its Port (0 beyond an open face, the Node's own level on a folded axis), the Node's own level now, minus its level before, plus the remainder kept at the Node, divided once by the wall w ; the quotient is the next level and the remainder stays. The remainder is the Node's own and never travels [1, the line].

At the vacuum's pace $p_0 = p_a = \Gamma$ the self coefficient vanishes and the line is the plain second-order wave rule of the pair,

$$a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a}), \quad (2)$$

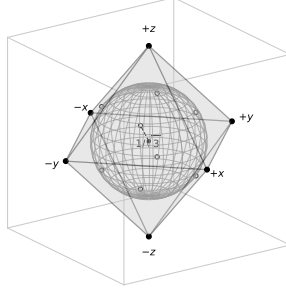


Fig. 2 The six Ports of one Node, $\pm x, \pm y, \pm z$, and the octahedron they span: the Nodes one interval away. The maps of the six Ports onto themselves that keep opposite Ports opposite are the 48 signed permutations of the axes, the symmetry group of the cube, under which the law is covariant. Drawn from the definitions; no run.

whose rotation at the wave vector $\mathbf{k}$ is $\cos \omega = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$: at wave number zero $\cos \omega_0 = \text{num}/\text{den}$, and for the massless pair $[1, 1]$ along one axis $\cos \omega = (2 + \cos k)/3$, so that $\omega = |k|/\sqrt{3}$ as $k \rightarrow 0$. Light's speed at long wavelength is therefore $1/\sqrt{3}$ Links per interval, under the causal bound of one Link per interval, the arrival's own speed; the two differ, and what in nature is one Link and one interval rests on the units of Section 8 [1, the lattice constants].

The clock once and the Link twice. From the coefficients, the rotation at a Node of the clock's pace p_0 and the Link's paces p_a is (DERIVED)

$$\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{ den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a), \quad (3)$$

the mass term scaling with the clock's square and the band term with the Link's pace. Two readings follow and no third [1, the clock once and the Link twice]. (i) *The clocks shift alike*: at $k = 0$ every massive band's rest rotation obeys $1 - \cos \omega_b = (p_0/\Gamma)^2 (1 - \cos \omega_0)$, one factor for every family, so two bound bodies at one Node shift their clocks alike, and a packet's fall in a pace gradient does not depend on its family beyond its band's own shape. (ii) *The Link is twice*: along a Link every wave's speed falls with $p_a = \Gamma - 2c - t_a$, the held level c entering the Link twice and the clock once, so light, which reads the held rows as every family does, is delayed and bent by twice the clock's share (Section 6). The line is not conformal: a light wave at a fixed wave number scales with $(p_a/\Gamma)^2$ and a bound mode at rest with $(p_0/\Gamma)^2$, and the two differ already at first order in c/Γ .

A held family's field steps at its row's pair with the pace 1 and the wall 3 den , at first order: for the massless row $a_{\text{next}} = (S_6(a_{\text{now}}) - 3 a_{\text{before}} + r) \text{ div } 3$, S_6 the sum over the six neighbours, and for a row with a gap 3 den $a_{\text{next}} + r' = \text{num } S_6(a_{\text{now}}) - 3 \text{ den } a_{\text{before}} + r$, the same call at the row's own pair [1, the vacuum's row steps at the divisor 1].

2.3 The direction

Let σ be the direction, $+1$ forward and -1 backward, let div and mod be the floor division and its remainder in $[0, w)$, and let ρ be the remainder carried in. With

$$u = \sigma \left(\sum_a R_a \text{arr}_a + S x - w y \right) + \rho, \quad z = \sigma(u \text{ div } w), \quad \rho' = u \text{ mod } w, \quad (4)$$

forward $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$ gives $(z, \rho') = (a_{\text{next}}, r')$, and backward $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$ gives $(z, \rho') = (a_{\text{before}}, r)$.

Theorem 1 (the direction). *The backward call returns (a_{before}, r) bit for bit; the reversibility of every step needs no second function.*

Proof. Forward, $\sigma = 1$, the line is the definition of div and mod . Backward, $\sigma = -1$: write Y for the right side's first two terms, $\sum_a R_a \text{arr}_a + S a_{\text{now}}$. Then $u = w a_{\text{next}} - Y + r'$ and $z = -\lfloor u/w \rfloor = \lceil (Y - w a_{\text{next}} - r')/w \rceil$. From the forward line, $Y - w a_{\text{next}} - r' = w a_{\text{before}} - r$ with $0 \leq r < w$, whose ceiling divided by w is a_{before} exactly; and $\rho' = u - w \lfloor u/w \rfloor = w a_{\text{before}} - (Y - w a_{\text{next}} - r') = r$. Checked bit for bit on 20,000 random cases against a forward and a backward function written separately [1, the direction]. $\square$

2.4 The four acts and what they can form

Rule3's line is one act; declared coefficients make it four: (a) the read, the line over the GameBoard with its six arrivals; (b) the one-Node step, the line with the neighbours' reads declared 0, $a_{\text{next}} + a_{\text{before}} = (a/b) a_{\text{now}}$, the rotation by the angle whose doubled cosine is a/b (Chebyshev's recurrence), which with $a/b = 1$ and the coefficient on a_{before} declared 0 keeps a level and with $a/b = 2$ from $(0, 1)$ counts; (c) the division, the line with $R = S = 0$, Euclid's division with the remainder kept; (d) the load, a declared integer in the numerator, how a count enters a level. A composition of the acts is a finite sequence of them, each with its coefficients declared in the files, applied to declared levels [1, the four acts].

Theorem 2 (what the acts can form). *Every composition of the four acts is a piecewise-linear function of the levels with integer slopes: a sum of linear forms and of floors of linear forms, composed. No such function is quadratic in the levels.*

Proof. Each act is a linear form of the levels or the floor of one; a floor of a linear form is piecewise linear with integer slopes; sums and compositions of such functions are of the same kind. The second difference of such a function in any one level is 0 almost everywhere, while the second difference of a product $\text{now}_i \text{before}_j$ in the pair $(\text{now}_i, \text{before}_j)$ is 1. So nothing bilinear comes from the acts alone. $\square$

What the theorem separates: every product of two levels in the engine is a *booking*, a reading of a bilinear form with declared coefficients and no write (Section 3.1); the one thing written from a product is the count, by the count's line, and that product is Rule3's own conserved current (Section 2.6).

The three tests and the six verbs. A line enters the law only if it is generic (one primitive with declared integers, no family's name and no kind), vector (one of the acts on the state, no root and no float) and local (its own record and the six neighbours, nothing kept at a Node beyond the law's numbers); a hypothesis that needs more is stated under its own name, outside the law. Every integer of the law is made of six operations only: the translation of an accumulator by its rate, the bilinear form with a declared matrix, the group-ring addition, the permutation, the evaluation at the primitive root of unity, and the Euclidean division with the remainder kept; a root, a true division or a draw is outside the law. No act of the interval takes a square root: the integer square root stays only at load and in the generator, as the fixed point of the division act iterated, $x \leftarrow (x + n \text{ div } x) \text{ div } 2$ from above until it repeats [1, the three tests of every rule][3, the six verbs; the root leaves everywhere].

2.5 The paces and the guard

The pace of a Node for a family is what its reads make of the Node clock:

$$p_0^2 = (\Gamma - c)^2 + c^2, \quad p_a = \Gamma - 2c - t_a, \quad c = \sum_{\text{reads}} \text{weight} \times \text{by} \times \text{the read family's time level}, \quad (5)$$

t_a the axis content from the read families' tensions on the axis a (each $\text{weight} \times \text{by} \times \text{the } aa \text{ part div } 2$, one division per read per axis, rounded at the read since the pace is a coefficient and no level). A positive weight is a hollow (the read family's level slows the clock, an attraction); a negative weight is a hill. By sign the weight is multiplied by the reader's q at the Node, the sign of its own rotation sense there (Section 3.4). A family with no reads steps at $p_0 = p_a = \Gamma$, the plain rule. The clock's slowing is this pace: a record at a Node of pace p rotates and moves as a record at Γ does, with its intervals p/Γ as long. Every family of quanta reads every holder of the content at the weight 1, a row without a gap and a row with a gap alike as its stepped field, $x = (L_b + L_g)/\Gamma$ with L_b the binding holder's level and L_g gravity's, and every holder of the sign by its own q ; a family sources what it reads at the weight it reads with [1, the paces; the reads are the held row's pair's].

The clock's second order is the composition. Each unit of the level slows the clock as it already stands, $p_0 = \Gamma(1 - 1/\Gamma)^c$, whose square to the second order is $(\Gamma - c)^2 + c^2 = \Gamma^2(1 - 2\Phi + 2\Phi^2)$ in the level's share $\Phi = c/\Gamma$; Rule3's coefficients take the paces' squares as written, with no division act, no rounding and no root, and $p_0 = \Gamma - c + c^2 \text{ div } 2\Gamma$ to the unit is a reading of the algebra; the Link's pace stays $\Gamma - 2c - t_a$, the level entering it twice [1, the clock's second order is the composition]. Section 6 reads the perihelion off this second order.

The guard, on squares, once at load. $0 < p_a$ and $p^2(\text{den} + \text{num}) \leq 2 \text{den} \Gamma^2$ at every Node, with the edge's square $P^2 = 2 \text{den} \Gamma^2 \text{ div } (\text{den} + \text{num})$, Γ^2 for light and above it where $\text{num} < \text{den}$, read on the clock's square and on each Link's pace alike over the whole initial state and never in the interval. Below 0 a pace is no clock; beyond P the step is unstable: the mode at wave number π on every axis has the factor $2 - 2(1 - \text{num}/\text{den})(p_0/\Gamma)^2 - 4(\text{num}/\text{den})(p_a/\Gamma)^2$, which is $-2 \text{num}/\text{den}$ in the vacuum and stays at or above -2 wherever p_0 and p_a are both at or below P , since $(P/\Gamma)^2(2 - 2 \text{num}/\text{den} + 4 \text{num}/\text{den}) \leq 4$ exactly when $P^2(\text{den} + \text{num}) \leq 2 \text{den} \Gamma^2$ (THEOREM). In the run no act reads a Node to refuse: every coefficient is a square, $R_a = 2 \text{num} p_a^2$ passes through 0 as the content reaches $\Gamma \text{ div } 2$ (the Link closes: the horizon) and reopens mirrored beyond it, and beyond P the mode at π grows until the host's trap on the arithmetic's width ends the run; the report gives the pace's minimum from the final state as a diagnostic [1, the guard][3, no one asks a Node; the root leaves the run].

The band at a pace. A reading of the line and no new rule: at a Node of paces p_0 and p_a a record's rotation at the wave number k along an axis is Eq. (3), so the band of rotations a body's Nodes carry narrows with the paces; a record whose rotation lies outside the band there is evanescent inside and reflected whole, and one whose rotation lies inside is slowed to the band's group velocity. Light, [1, 1], meets a total mirror where $p_a < \Gamma \sin(k/2)$, and a window otherwise [1, the band at a pace]. At a Node of content $x = L/\Gamma$ light's line has $S/w = 2 - 2(1 - 2x)^2$ and $R_a/w = (1 - 2x)^2/3$, its band is $[2 - 4(1 - 2x)^2, 2]$, and at long wavelength $\omega = (1 - 2x) |k|/\sqrt{3}$: light's speed is $c_0(1 - 2x)$, an index $n = 1/(1 - 2x)$, $n - 1 = 2x$, the content read twice (Fig. 3) [1, the rows against nature, (b)]. At $x = 0$ light's zone-corner mode sits at -2 with no margin, and at any negative content it grows, $\cosh \mu = 2(1 - 2x)^2 - 1$ with μ the growth per interval; the vacuum content of Section 4.5 gives it the margin $8x_{\text{vac}}$ [1, what is open, item 22].

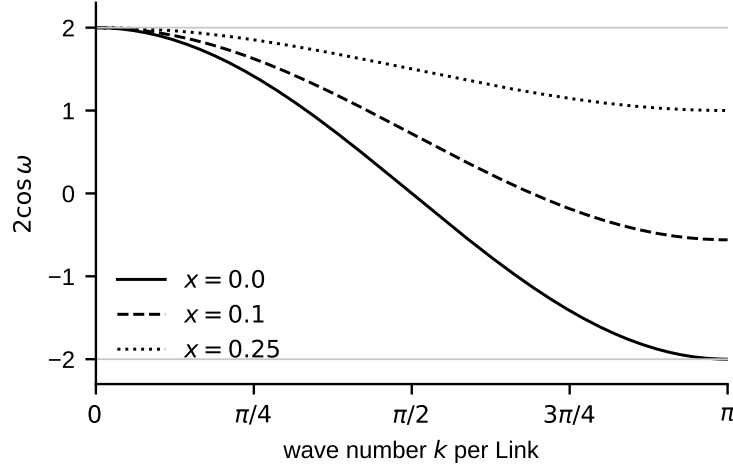


Fig. 3 Light's band at a content $x = L/\Gamma$ read into the paces (DERIVED, Eq. (3) at num = den): the doubled cosine of the rotation along one axis against the wave number, $2 \cos \omega = 2 - 2(1 - 2x)^2(1 - \cos k)$, at $x = 0, 0.1$ and 0.25 . The band's bottom rises with the content, its top stays at 2; at the horizon $x = 1/2$ the band closes to a point. The slope at $k = 0$ is light's speed, $(1 - 2x)/\sqrt{3}$ Links per interval. Drawn from the definitions; no run.

The lattice constants. The six Ports' own, exact and no number of nature: the massless row's rest around one static source of s quanta per interval at the divisor 1, by the start's iteration on an open box with its faces' constant image removed, is $3G(r)s$ with $3G(0) = 0.7582$ at the Node and $0.2582, 0.1286, 0.0826, 0.0608$ and 0.0483 at 1 to 5 Links, toward $3/(4\pi) = 0.2387$ far away (COMPUTED); of which $0.76s$ at the Node and $0.24s/r$ beyond are the continuum's; and light's $1/\sqrt{3}$ [1, the lattice constants].

2.6 The conserved form

The read matrix $2 \text{ num } p_i^2$ is not symmetric where the paces differ, so the form carries the weight $1/p_i^2$ on the Node terms and the plain current on the Links:

$$E = \sum_i \frac{w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{ now}_i \text{ before}_i}{p_i^2} - 2 \text{ num } \sum_i \text{ now}_i S_6(\text{before})_i. \quad (6)$$

Theorem 3 (the conserved form). *At constant paces E is exact under the line without the rounding; with the rounding each Node's remainder adds at most one unit of the level per interval to the walk of E . Its Link term, the plain current $\text{num}(\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j)$ through the Link ij , is the current of Rule3's own conservation law.*

Proof. Without the rounding the line is $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$ with $\mathbf{M}$ the matrix whose diagonal is S_i/w and whose entries on the six Links are R/w . Let $\mathbf{D}$ be the diagonal matrix of the weights $1/p_i^2$; then $\mathbf{DM}$ is symmetric, since $R_a = 2 \text{ num } p_a^2$ at the reading Node and the weight $1/p_i^2$ cancels it on both sides of every Link. For a symmetric $\mathbf{K} = \mathbf{DM}$ the quantity $Q(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{D} \mathbf{x} + \mathbf{y} \cdot \mathbf{D} \mathbf{y} - \mathbf{x} \cdot \mathbf{K} \mathbf{y}$ satisfies $Q(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = Q(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}})$: with $\mathbf{a}_{\text{next}} = \mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}$, $\mathbf{a}_{\text{next}} \cdot \mathbf{D} \mathbf{a}_{\text{next}} - \mathbf{a}_{\text{next}} \cdot \mathbf{K} \mathbf{a}_{\text{now}} = \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{a}_{\text{before}} - \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{M} \mathbf{a}_{\text{now}}$, which is the other side. Multiplied by w this is E up to the Link term's sign convention [1, the conserved form]. $\square$

E is a GameBoard reading. It changes by the work term where the paces move: the total form $D = |\mathbf{a}_{\text{now}}|^2 - \langle \mathbf{a}_{\text{next}}, \mathbf{a}_{\text{before}} \rangle$ of a record under paces that change in time obeys exactly $D(t+1) - D(t) = \langle \mathbf{a}_{\text{next}}, (\mathbf{M}(t) - \mathbf{M}(t+1)) \mathbf{a}_{\text{now}} \rangle$ (THEOREM, by substitution), 0 at fixed paces and negative where a well deepens; the count's line, moved by the flux alone, keeps the count meanwhile: the mass defect is the form's fall at a kept count (Section 5) [1, the mass defect is the form's fall at a kept count].

2.7 The tension: the momentum's flux is the wave's own stress

The momentum density along the axis a at the Node i is the count's line's current per axis, $P_a(i) = (F_{i,i+a} - F_{i,i-a})/\text{num}$, and Rule3's line gives exactly, at constant paces (THEOREM),

$$w [P_a(t+1) - P_a(t)] = \sum_b R_b [G_{ab}(i-b) - G_{ab}(i)], \quad (7)$$

$$G_{ab}(i) = \text{now}_i (\text{now}_{i+a+b} - \text{now}_{i-a+b}) - \text{now}_{i+b} (\text{now}_{i+a} - \text{now}_{i-a}),$$

$G_{ab}(i)$ the stress booking on the b -Link $(i, i+b)$ at one time, the levels now alone; the proof substitutes $w \text{ now}' = \sum_b R_b (\text{now}_{i+b} + \text{now}_{i-b}) + S \text{ now}_i - w \text{ before}_i$ into $w P_a(t+1)$: the S term cancels, the w term is $w P_a(t)$, and the R term is the difference of the fluxes. So the flux of the a -momentum across the b -Link is $-(R_b/w) G_{ab}$, the second booking of the conserved form (its Link term shifted by one Link), and the tension on the axis a at the Node, the part Rule3 reads into the Link's pace and the source of the held row's axis part aa , is the momentum's flux, the mean over the Node's two a -Links,

$$T_{aa}(i) = -\text{num} (G_{aa}(i) + G_{aa}(i-a)) \text{ div } 2, \quad (8)$$

$$G_{aa}(i) = \text{now}_i (\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a} (\text{now}_{i+a} - \text{now}_{i-a}),$$

the neighbour's own difference brought through the Port and no other shift: the tension on a Link is the booking of its two ends' own readings, a level and a difference each, and a Node sources its tensions with half of each of its Links, so the line is local as F is. With the rounding the identity holds up to the remainders' term, $-r'_i \Delta_a \text{now}_i + \text{now}_i (r'_{i+a} - r'_{i-a})$, exactly. It is read and never written, quadratic in the amplitude as the count is: a plane wave of amplitude b at the wave number k has $G_{ab} = -2b^2 \sin k_a \sin k_b$ exactly, so the flux along the wave is positive, as a wave's pressure is; a uniform record gives 0; a standing body's tail carries a tension that does not vanish in the mean; for a packet of count c and velocity $\mathbf{v}$ it is $c v_a v_b$ within the rounding, the dust's stress with no count in any divisor and nothing undefined at a count 0. With the sign as written a wave's stress deepens the content around it: light gravitates by its pressure too. The off-diagonal parts and the vector part, which Rule3 never reads, are not written [1, the tension; the final unification][3, the tensor part is the wave's own stress].

2.8 The one write

Every write into a family's level on the GameBoard is one act, the third line of Eq. (1): $(w q(\text{Node}) + r) \text{ div } E$ at a Node each interval, the division act with the remainder r carried at that Node; q the writer's own quantity there (a family of quanta's well $D_i \text{ div } T$, its Wronskian's quanta, or its tension T_{aa}), w the one weight of the pair of families (whoever reads with w sources with w), E the written row's divisor (E_s ; $E_s W_c$ for a tension, one act per axis). The hold, the well and the start are its instances; no number is the act's own. Generic (the rows' keys), vector (one carried division), local (the Node) [1, the primitives][3, the one algebra]. The form $D_i = \text{now}_i^2 - \text{next}_i \text{ before}_i$ of a

record at a Node is its well's source: for a standing record it is $B^2 \sin^2 \omega_b$ at every interval, B its peak [1, the gate reads the form at any phase].

2.9 The interval and its inverse

The engine steps every family from the state at the interval's start to the state at its end; every read is of the start's values, one Link per interval and nothing in zero time. Before the first interval the whole initial state is checked once against the guard; then the interval has four acts, in order: (i) the signed read, every family's paces from the held families' levels at the start; (ii) Rule3 on every record, each family of quanta's two level pairs by its pair and its paces, each holder of the content's parts at its own pair at the pace 1; (iii) the count's line and the sense's line on every family's count and sense, the tension on each axis booked beside them, and the clicks at the Nodes declared to report; (iv) the hold, each held family's time part gaining its source over its divisor and its tensions the sources' momentum fluxes; light is born here, by the write of a rotating body's varying Wronskian into the sign holder's record [1, the interval][2, the interval].

Backward, the acts run by Rule3's direction -1 : a held row's hold back once every family that sources it is booked back, a family of quanta booked back once its own hold is off, so the holder of the sign goes before the rows it sources; every interval returns bit for bit, since every level a family read at the interval's start is one the state after the interval still holds, and only the lay of the first act is not taken back. The back-in-time gate is the whole run: N intervals forward and N back, every array of every family compared bit for bit at every interval on the way back, MATCH or MISS naming the first interval and the first Node that differ; a look is not run before it says MATCH on the look's own world (THEOREM by construction, from Theorem 1 and the count's line's inverse, Section 3.2). On the two slits' worlds of Section 7 it says MATCH over their 130 intervals [1, the interval; what is open, item 21][3, the board goes back in time, as a gate].

2.10 The three channels

One family acts on another in three ways and no fourth (Fig. 4): *the read*, at (i), a held row's level times the read's weight into the reader's paces, the clock once and the Link twice, multiplicative, the whole of gravity, of binding and of the charge's pull; *the write*, at (iv), the writer's own quantity through the one weight and the written row's divisor into the held row's level, additive, the source of every field and the birth of light; and *the click*, at (iii), the count's line's move of one whole quantum, the measurement alone. No product of two levels is written; only the current is read and the count written from it. No bilinear coupling between families is in the law: the world of clicks (bodies, detectors, the counts) and the world of Nodes (the records) run as one loop and touch only at the click and at a body's Nodes; a record sources the families its own family reads, so every pair of families that read each other is reciprocal, action and reaction one key [3, the coupling is the click alone; light is born by the write].

3 The click

The click is one mechanism: the count's line carries every family's quanta, and a Node declared to report reads each quantum that arrives whole [1, the primitives]. This section states the booking, the count's line with its theorem, what the click ends (nothing), the sign, the birth of light, absorption, the price of locality and the postulates.

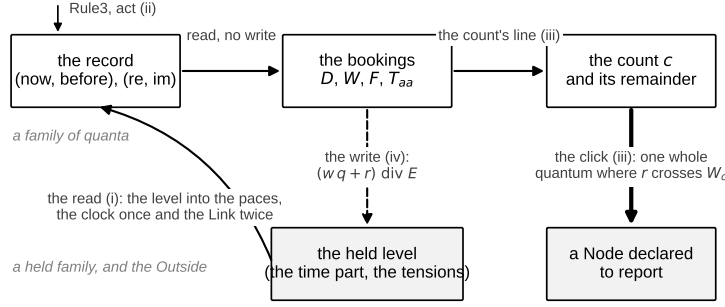


Fig. 4 The three channels through which a family acts on another, drawn from the interval's four acts (Section 2.9). A family of quanta steps its record by Rule3 and books its form and its current; a held family carries a level. The read (i) takes the held level into the record's paces; the write (iv) takes the record's form, its Wronskian's quanta and its tension into the held level over the row's divisor; the click (iii) moves one whole quantum of the count where the remainder crosses the wall, and a Node declared to report reports it. Every arrow is one line of Eq. (1). Drawn from the definitions; no run.

3.1 The booking

The current of a record into the Node i from its neighbour j through their Link over an interval is the bilinear form of its two levels at the two ends, $F_{ij} = \text{num}(\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j)$, positive inward: a record of wave number k travelling from i to j has $F_{ij} = -2 \text{ num} |\psi|^2 \sin \omega \sin k$, below 0. It is antisymmetric, $F_{ji} = -F_{ij}$, and it is the Link term of the conserved form (Theorem 3). The booking reads it at a Port and writes nothing [1, the booking].

3.2 The count's line

Every family of quanta carries a count c , the number of its quanta at a Node, one count per family over the GameBoard; its wall is $W_c = 3 \text{ den } T$, T the quantum's action; its read is the current. At every Node the second line of Eq. (1) holds: Rule3 with the coefficient 1 on each axis's inflow through its two Ports, W_c on the count and the wall W_c . The click is the division: a quantum moves whole from one Node to the next exactly when the remainder crosses W_c , and never a fraction of one. The inverse is the same line with the current reversed, $W_c c_{\text{now}} + r = W_c c_{\text{next}} + r' - \sum_j F_{ij}$, exact, the currents read from the interval's levels as the forward step read them. The line has no other act: a total below zero gives the count -1 with the remainder in range, the hole, which the line conserves and fills; a clamp at 0 would create quanta, and a block of a Node's currents would be a branch on a Node's value that loses the inverse, so neither is a rule [1, the count's line; the inverse][3, nature pushes toward whole numbers].

Theorem 4 (the count is conserved). *Over any region of Nodes, $\sum (W_c c_i + r_i)$ changes in one interval only by the currents through the region's boundary Links.*

Proof. Summing the line over the region, each interior Link's current enters one Node's right side as $+F$ and its neighbour's as $-F$ and cancels; what remains is the sum over the boundary Links, a telescoping sum, exact in integers. On a periodic GameBoard with no face the sum is constant to the bit [1, the count's line]. $\square$

The remainder's origin, and the lay. The remainder at a Node starts at the origin $W_c \text{ div } 2$: started at 0, a Node with no quantum reads the count -1 at its first outward swing. The count is laid once, at the first act, from the weighted share, the conserved form's Node term over the Link's pace squared less the plain Link term, $e_i = [w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i] \text{div } (2p_i^2) - \text{num now}_i S_6(\text{before})_i$ with p_i the Link's pace at the Node, in the current's units, $W_c c + r = e_i + W_c \text{div } 2$; its change over one interval of Rule3 is exactly $\sum_j F_{ij}$ at every Node, in the vacuum and in a well alike, so the count's line is the continuity of Rule3's conserved form everywhere (THEOREM). After the lay the count moves by the line's flux alone. In the vacuum the share is the plain share, whose sum over a closed GameBoard is $3 \text{ den } \sum_i D_i$; in a well the plain share is the local form D that sources the fields and no count, so a body in its well is lighter in every field it sources than in quanta [1, the remainder's origin; the lay and the wall][3, the lay from the weighted share].

The count's unit. The engine's count, the weighted share at the Node's own paces, is the count the law conserves; a count in the vacuum's share (the plain share at the vacuum's coefficients) counts the same record less at a deep Node, 3.7 times less at the Node of the 2,000 pixel of Section 5 and 3.1 times less in all, and the same to a percent at a cloud, whose content is 0.005. So a world is declared at the engine's count, the unit-free numbers stand, and the counts of the compact branch are restated by the generator's conversion at the Node's paces [1, the count's unit].

The velocity. The record's wave number is conserved by the record's line where no pace gradient stands (Rule3 is translation-invariant), and the count's line is the continuity equation of the current in integers, so the count's centroid moves at the current's group velocity, with no leak beyond the remainder; a body at rest has zero net current at every Node in the mean and its count stays in place; a moving body's clicks keep its velocity [1, the velocity].

3.3 The click ends nothing

Assumption 1 (the click ends nothing). The click is the count's line's move of one whole quantum where the remainder crosses the wall, at one Link, local; the record goes on after it and nothing ends anywhere. The count that arrives at a detector's Node is the detector's click, a detector's count is a family's count at its Nodes, and a detector reports and takes nothing. The ending of a record at a click by the detector, a Node that takes the quantum into its own count, is decided against and is no hypothesis of the law.

This is the owner's decision of 2026-09-30 [3, the click ends nothing], and it is the law's claim to be tested against nature, not a deficiency: perhaps nature is local. What follows from it needs no further act. One quantum is reported once and never twice, by the count's conservation (Theorem 4) and by no act at a distance; no signalling holds trivially, since nothing at one detector changes anything at another; the Outside reads the clicks alone; and the click is the one reading at a body's scale. A click has a family and an axis: the family whose count moved and the Port the quantum crossed, its axis and side, the count's line's tally; the three tallies are what a detector reads of the click's axis, no charge of any body and no force of their own [1, the count's line; the third and the colours, closed as absent].

3.4 The sign is the rotation sense

A family of quanta carries its record as two level pairs, (re, im), two solutions of the same line stepped by one call of Rule3 with one set of coefficients, each with its own remainder; a held family's parts stay real. The charge density at a Node is the booking

$$W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}}, \quad (9)$$

the Wronskian of the two pairs, which Rule3 conserves exactly with the conserved form's weights $1/p_i^2$ (THEOREM, **DM** symmetric), its local change the Link current num $(\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$ at the levels the step started from; the sense is laid once at the first act from $w W_i \text{div} (2p_i^2)$ at the count's wall and moved by the count's line on that current, $\sum (W_c k + r)$ conserved to the bit. A real record $(\text{im} = 0)$ has $W = 0$, so light, the sign holder's own real record, is neutral; a rotating record has W of either sign by its sense, $-A^2 \sin \omega$ for $e^{+i\omega t}$ and $+A^2 \sin \omega$ for $e^{-i\omega t}$. The reader's q at a Node is the sign of its own sense there; the holder of the sign is sourced by the Wronskian's quanta, $(W_i + r) \text{div} T$ each interval, over its divisor; pair creation is the conversion between the two senses; the coupling goes as the sign, never as Q/M , so an electron and a proton accelerate alike in one field [1, the sign is the rotation sense].

The charge is the record's, not the count's. For a record $(\text{re}, \text{im}) = (A \cos \omega t, eA \sin \omega t)$ the booking is $W = -eA^2 \sin \omega$: the charge is the count's size times the ellipticity e of the rotation, a property of the record that Rule3 conserves (the two levels obey one linear line, so their ratio is carried, not written). A circular quantum ($e = 1$) has $|W|/D = 1/\sin \omega$; a nearly real cloud of 1,836 quanta with $e = 1/1,836$ has the charge of one circular quantum; a body is one mode with one sense and one ellipticity at all its Nodes. What it costs, a finding against nature: nothing in the law quantises e , so a body's charge is continuous, and $|q_p| = |q_e|$, nature's to 10^{-21} , is an input and not derived; the line that would derive it is a hypothesis under its own name, *the second level's amplitude is a count of its own* (Section 8.4) [1, the sign is the rotation sense; the hypotheses under their own names].

3.5 Light is born by the write

Light is the sign holder's own record: the holder of the sign is one record whose time part is its real level pair, stepped by Rule3 at the paces of its reads (the content, so light is bent) and written by the hold from the Wronskian's quanta of the families that read it by sign. A static rotating body writes a static level (its well in the sign holder, Coulomb's) and radiates nothing; a breathing or moving body writes a varying source at its Nodes, the row carries it away at one Link per interval in the vacuum, and that wave is the light (DERIVED). Its count needs no lay and no conversion: every family runs the count's line on its own record, a held family's count starting at 0 everywhere where its record is 0 and moved by its currents, so its total stays 0 within the remainders' walk, quanta and holes -1 together, the line's own pair creation: a photon born with its hole. A body's quanta that leave it stay its family, messages of matter, and the matter count is conserved by the line to the bit; no quantum changes family at a crossing, no ledger of bodies stands beside the Nodes, and a body's Nodes are where its family's count stands about its declared Nodes, derived when a report needs them and never kept. A body reads the sign holder by its own sense at its own Nodes, so a breathing body reads the light it writes and its pace answers, the radiation reaction, in the same loop [1, light is born by the write][3, light is born by the write; the coupling is the click alone].

3.6 Absorption is the read by the sense and the write

Assumption 2 (absorption). A family of quanta reads the sign holder's level by its sense, so the light's level at a body's Nodes enters the body's paces, a drive at the light's frequency, resonant where it meets the body's rotation (the parametric line, at twice the rotation for a modulated pace), and the body's write answers, the driven body writing light of its own; behind the body the written wave and the incoming one cancel where they are out of phase; where the wave is cancelled its current stops, so the light's count that arrived at the body's Nodes does not leave, the count's line piles it there as light's count with no wave to carry it, and a Node told to report reports the rise. A

light quantum never becomes a matter quantum; the conversion act is refused as unnecessary and as a branch, and matter's count stays the baryon number.

This is the owner's decision of the same day on the advisor's line [3, the click ends nothing][1, the count's line]; it is a body's act and no new act of the law: the read (i) and the write (iv) with the sense in them, and the count's line as it is. So feeble light and a lone quantum spread over many Nodes never click at a cold detector Node, every remainder 0 (they move no remainder over W_c there), and at a resonant absorber one quantum clicks once, locally: a body that absorbs the whole wave over its resonance time integrates the wave's current into its Nodes' remainders, whose sum is the wave's count, one quantum, crossing once. Nature's detector is a body and not a Node. The free electron's detection is the same with matter's line, and the radiation reaction is the same loop. An absorber in one slit takes that slit's wave and the screen shows the single slit's pattern, nature's loss of the fringes by a detector in the path with no act at a distance; a bare detector Node in a slit changes nothing, and a marker that does not absorb has no line. What the absorption look reads, computed with the read alone before the write is built, is Section 7.5.

3.7 The price of locality

The law's word against nature, decided and against nature until tested [1, what is open, item 4]:

- (i) *The single-quantum statistics, and the two takers.* A detector's sensitivity is its remainder: a quantum moves only where a remainder crosses the wall W_c , so a dilute wave that brings a detector Node the fraction f of a quantum carries it over when the Node's remainder already stood within f of the wall, a whole quantum rising there, taken from the neighbours, whose remainders go negative by the same, local, exact and conserved. On a cold board, every remainder 0 at the start, a detector rises only on a whole quantum, so a dilute packet reports nothing. On a warm board, the remainders at the detectors spread evenly over $[0, W_c)$ as many passages of light leave them, the chance that a detector rises on a dilute quantum is the fraction of the quantum's count that reaches it, $P(\text{click at } i) = f_i$, the intensity's share, and the mean number of rises per quantum is $\sum_i f_i = 1$ exactly by Theorem 4: Born's rule for the detectors from the count's line with no probability in the law, the randomness the remainders' history, deterministic and unreadable from outside (DERIVED, the advisor's line, no line added). What stays against nature, with a number: two far detectors do not know of one another, so on one quantum split f_1 to one and f_2 to the other the law gives both rising with the probability $f_1 f_2$ (a quarter at an even split), neither with $(1 - f_1)(1 - f_2)$ and exactly one with the rest, two rises leaving two holes behind, where nature gives never both: the law's claim, the price of the locality the owner chose, the same price as Bell's, in a number a look can read, a beam split evenly onto two warm detector groups and the rate of double rises against a quarter [1, what is open, item 4][3, the click ends nothing].
- (ii) *The cold detector Node.* A lone quantum spread over many Nodes never clicks at a cold detector Node, and a reporting detector Node in one gap of the two slits changes nothing of the screen's row, since nothing absorbs (the run of Section 7 shows both); nature's which-way loss of the fringes needs an absorber, which the law has as a body (Assumption 2) and not as a Node.
- (iii) *Bell's row.* Two labels of one written wave, each clicking alone at its own end: the pair's birth, the turn at each end and the labels' clicks are open under the count's line; the blind expectation stands, $E(a, b) = \cos 2(a - b)$ with each side's marginal one half and $S = 2\sqrt{2}$ at the settings 0, $\pi/4$, $\pi/8$, $3\pi/8$, and nothing here derives it [1, the rows against nature, (h)].

The two slits' dilution series on a warm board and the absorption look test (i) and (ii), and their result is the law's answer, not a line to add. The paper writes these findings as they stand.

3.8 The postulates

1. The world is Nodes and Events. Space is the GameBoard; an Event is a wall crossed by a remainder at a Node: a level stepped, a count moved, a click; nothing else happens.
 2. A Node holds bounded local information: its NodeState, fixed in size, read from itself and its six neighbours; no read beyond them, no list that grows with the world, nothing of the past kept. A self-field estimated or subtracted from anything global is forbidden, whatever the size of its final answer.
 3. Consistency is local and causal: an Event first changes its own Node and travels Link by Link, one Link per interval, each Node updating on receipt under the same rule; there is no exception, and no quantum changes family at a crossing.
 4. The causal speed is one Link per interval, built in; every other speed is a rational, a count of Links over a count of intervals.
 5. Every physical calculation is on bounded integers; there is no float, no root and no draw in the law; a Node keeps only the law's own numbers.
 6. A measurement is a detector's click, an action of the law on the state: a count arrives at the detector's Node and the detector's count changes. Only a click is compared with nature or pinned as an expectation; displays and the host's readings read state and write nothing.
 7. The Inside is the GameBoard, where no one measures; the Outside is the detectors and their clicks, the only thing claimed to represent nature.
- [1, the postulates]

4 The families from the rule

A family is a band of the one rule and not a declaration, and its row holds nothing the rule and the geometry fix: its pair $[\text{num}, \text{den}]$, $\cos \omega_0 = \text{num}/\text{den}$, and what it holds, the content or the sign, at a divisor. Every pair is written over the GameBoard's one denominator Γ , so the file holds the numerator m alone, $\cos \omega_0 = m/\Gamma$, and the wall is $w = 6\Gamma^3$ for every family; a reduction to lowest terms is a reading and never the loader's act, so every family's remainder has the one resolution [1, the families from the rule; the mass is one integer; the wall is one].

4.1 The exact bands, and the third closed as absent

The line of one Node at rest, $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$, is exact in integers only where $2 \cos \omega_0$ is an integer, so the rule carries exactly three exact rotations, $2 \cos \omega_0 = 1, 0$ and -1 , the periods 6, 4 and 3, the pairs $[1, 2]$, $[0, 1]$ and $[-1, 2]$ (THEOREM). A record on an exact band at rest at the vacuum's pace leaves no remainder in its rest rotation, so it never clicks and only turns: no remainder, no click. The exact bands are exact only in the vacuum: at a Node of pace p_0 the band's rest rotation is $2 \cos \omega = 2 - 2(\text{den} - \text{num})(p_0/\Gamma)^2$, an integer at $p_0 = \Gamma$ alone, so a record on an exact band inside a well leaves a remainder and clicks like every family's [1, the exact bands; the exact bands are exact only in the vacuum].

The third, $[-1, 2]$, is a band the rule allows and the universe does not use, the mirror of $[1, 2]$: one family on it holds no standing body (its read is negative, so its own well pushes its rest rotation up into the vacuum's band and the record leaks). The three colours are the count's line's three tallies, a detector's reading of the click's axis, no charge of the body and no force of their own; whiteness is their balance at rest. So the quarks are not in the law: the white body, the tube, the charges $2/3$ and $1/3$, the generations, up and down leave with the third, and the band $[0, 1]$ of the

period 4 is the same, a band allowed and unused [1, the third and the colours, closed as absent][3, the colours are the three axes].

4.2 The rule's own universe

The universe is the one whose every number is the rule's: the pairs from the integers 1, 2 and 3 of the three exact rotations and the three axes, every divisor 1 but gravity's, and the free numbers named. Its rows, as the file `examples/events/rule.json` holds them and as the rule derives the rest from the two keys of a row, its pair and what it holds (DERIVED) [1, the families from the rule; the rule's own universe][2, the loader and the files]:

Family, its rank	Pair	Holds, at the divisor	Parts	Clicks	Its free number
gravity, the time part and the three tensions	[1, 1] ([6000, 6000])	the content, E_g (1,000 in the looks)	1 + 3	never: a field	E_g , the hierarchy, and its rest c_{vac}
the binding holder, the strong force	[2400, 2401]	the content, 1	1	never: a field; its own waves have the gap $\cos \omega = \text{num}_b / \text{den}_b$	the pair, its range $R = 20$ Links, tied to Γ by the proton
the charge, its quanta light	[1, 1]	the sign, 1	1	its count by its own count's line from 0; light born by the write	none
the bound charge (the polarisation), the exact band	[1, 2] ([3000, 6000])	the content, 1	1	never	none; a field of under one Link's reach
matter, the scalar	[m , Γ] ([4000, 6000])	nothing	1	its count by the count's line, a quantum that leaves a body a message of matter	m , the mass

The universe's integers: the Node clock $\Gamma = 6,000$, a multiple of 6 so that every pair is exact over it; the quantum's action $T = 2^{15}$; the integers' width 63 bits, a bound and no physics; and the vacuum content c_{vac} (Section 4.5). Every family of quanta reads every holder of the content at the weight 1 and the holder of the sign by its own q , and sources what it reads at the same weight; matter reads gravity and the binding holder plainly, $x = (L_b + L_g)/\Gamma$, the bound charge, and the charge by its sign; light reads the same holders of the content and is bent by them. The three of Rule3 is the three axes; the three axes give the three tensions the massless row carries beside its time part, 1 + 3, the parts Rule3 reads and no other; every other family is a scalar. So the files hold Γ , T , the matter pair's m , the binding holder's pair, gravity's E_g and c_{vac} , every other divisor 1, and nothing else of physics; Γ and T are units, as c and $\hbar$ are in nature, and no line of Rule3 fixes T , the rule being linear and its form's scale free. The crack, named: a second massive scalar family, a second mass, the rule does not forbid, and nature's spectrum of masses is the open row under this line [1, the final unification][3, two units and the bounds; the standing goal].

Every row of the file is a record stepped by Rule3 at its pair and sourced by the hold's division act, the vacuum's row and every holder with a gap alike; no laid row, no fixed point computed

outside Rule3 during the run and no act beside the six verbs and the division act remains in the law or the engine; the start's rest is the fixed point of the division act with its remainder, the state Rule3 holds static, computed once by the generator (ASSUMPTION, the owner's word) [3, nothing that is not Rule3 remains].

4.3 The reach of a held family is its pair

A held family's rest solves $\text{num } \Delta^2 a - 6(\text{den} - \text{num}) a = -3 \text{den } \sigma$ (its line with the source once an interval, Δ^2 the six-neighbour second difference), so outside a body its rest falls by $e^{-\kappa}$ per Link with $\cosh \kappa = 3 \text{den}/\text{num} - 2$, the family's own band at zero rotation: the gap is the mass of the records and the inverse reach of the field, one number (DERIVED). [1, 1] has no gap and its rest is Poisson's over the whole GameBoard; [1, 2] has $\cosh \kappa = 4$ and falls by 0.127 per Link, a contact well of 0.41 Links' reach; and a pair with num just under den has $\kappa^2 = 6 \text{den}/\text{num} - 6$ to the second order, a range $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$ Links [1, the well].

4.4 The two rows: the binding holder and gravity under a divisor

Assumption 3 (the two rows, decided with the caveat). The well of a record, $D_i \text{ div } T$, is laid into two holders. The binding holder, a family with a gapped pair $[\text{num}_b, \text{den}_b]$, $\text{num}_b < \text{den}_b$, at the divisor 1, stepped by Rule3 at its pair as the vacuum's row is and started at its rest, which with the gap is a sum with a range, $R = \sqrt{\text{num}_b/(6(\text{den}_b - \text{num}_b))}$ Links and $e^{-r/R}/r$ beyond it: it binds the bodies, it is the law's strong force; its own waves have the gap $\cos \omega = \text{num}_b/\text{den}_b$, a quantum with a mass whose Compton length is R , Yukawa's own line and no input. The massless holder [1, 1] at the divisor E_g declared by the file: gravity, long-ranged and weak by E_g . A family reads both levels into its pace at the weight 1.

The first pair is [2400, 2401], $R = 20$ Links, $m^2 = 0.0025$, the quantum's rest frequency 0.029 per interval, the pair tied to Γ by the proton: a cloud of 1,836 quanta is bound under $R = 20$ and not under $R = 10$ or 15 (Section 5.3). In the looks E_g is the run's own number, large enough that gravity does not enter a body's binding (1,000 suffices at the pixel, 903/1,000 under one level); nature's E_g is about 3×10^{42} , 10^{36} times 1,836², and the hierarchy is the file's one number, as G is nature's. The reservation, in the owner's words, we may have been wrong: the line stands as a decision with a reservation and reopens on any of (i) the three tests, (ii) the bodies not standing under the screened sum, (iii) the charged cloud look reading electric over gravitational between two bodies as $E_g/1,836^2$ and nature's other way, (iv) the finding of scale below, which this line does not touch and which may be the deeper error [3, the two rows][1, what is open, item 20].

The road not taken, kept as the law's word. Under every divisor 1 the force that binds a body and the force between bodies are one row, the well read at the weight 1; the gravity between two quanta is then of order one against nature's 5.9×10^{-39} for the proton, and the electric force between two protons, a charged cloud being a nearly real record with one unit of W among 1,836 quanta (Section 3.4), is $1/1,836^2$ of the gravity between them, where nature's is 10^{36} times larger. That is the price of every divisor 1 with its number, written as the law's word on its own universe and its largest number against nature [1, the rows against nature, (v)]. *The finding of scale*, the same statement with no number: matter's rotation at rest, $\cos \omega_0 = 2/3$, fixes the electron's reduced Compton length in the lattice's units, light's speed over the rotation, $0.577/0.841 = 0.69$ Links; nature's proton is $1/400$ of that, 0.0015 Links, while the law's cloud of 1,836 quanta has the radius $2\Gamma/C$, at least 6.5 Links at $\Gamma = 6,000$ and larger at every larger Γ : nature's proton is bound thousands of times smaller than the well binds anything, so the well row binds too strongly at a

distance and too weakly up close to be both the proton's binding and gravity [1, the rows against nature, (v) (2)].

4.5 The vacuum content

Assumption 4 (the vacuum content). The massless row rests at a declared level c_{vac} everywhere, the vacuum content, a number of the file beside Γ and T , about 1 percent of Γ (60 levels at $\Gamma = 6,000$, the smallest that keeps every content of the look above 0), written as the law's cosmological constant, positive as nature's.

A constant is the massless row's own rest and its step keeps it ($a_{\text{next}} = S_6 \text{ div } 3 - a_{\text{before}}$ returns it); every reader's paces are $\Gamma - 2c_{\text{vac}}$, the same physics at an effective Γ ; light's zone-corner mode gains the margin $8x_{\text{vac}}$ in $\cos \omega$, $x_{\text{vac}} = c_{\text{vac}}/\Gamma$. The finding that led to it: a holder with a gap sourced by a changing well rings with its own waves, the law's mesons, above the gap travelling (0.029 per interval for [2400, 2401], 1.047 for [3000, 6000]) and at $k = 0$ standing where born, radiated by a breathing body as nature's nuclei radiate pions; its troughs below 0 far from the body are a finding, and light's refusal there is right, a real runaway of Rule3 (Section 2.5), where nature has no such runaway since its content is an energy and never negative. Beyond an open face, and beyond an inner face, a held row with a declared rest reads its rest, the same vacuum on both sides, so the massless row is c_{vac} up to the face and beyond it and light keeps its margin there, its waves seeing 0 deviation beyond the face and reflecting as before; a row without a declared rest reads 0 beyond. The other line, the vacuum marginal and every negative content ending the run, was not taken [1, what is open, item 22; the objects][3, the vacuum content].

5 The bodies

5.1 What a body is

A body is its count and its record over its Nodes, and never one Node: every body is many Nodes, and nothing gathers it. The count c_i is its family's count at the body's Nodes, in quanta; it enters Rule3 through the pace alone, $x = (L_b + L_g)/\Gamma$, and it never splits: a quantum moves whole, by the count's line. The record of the body's family splits by the read and is bound by the wells its counts source: by the binding holder's screened sum with the range R , while gravity's $1/r$ is weak by E_g and enters no body's binding [1, what a body is]. No law is written for a body: the one rule is written for the Node, and a body is the support of a mode of that rule; every proposal is a row at a Node or it leaves or is derived. A body has no law of motion of its own: its record moves by Rule3 alone, the count follows by the count's line, the momentum is a reading of the record's current, and in a pace gradient its packet accelerates as the mode's top velocity times its redshift per level times the level's difference across it [3, what a body is].

The four lines of the body (DERIVED from the line, with the decisions named): (a) *the body is the fixed point of its binding row*: its form sources every held row its family sources, its record is the standing record Rule3 makes in those paces, its count the weighted share that record lays, and it stands where the sources return themselves; (b) *the form sources the fields*: a held row's level is the rest the start solves from the bodies' forms at the row's divisor, then Rule3 steps it; (c) *the count moves by the count's line alone*: laid once at the first act and moved by the flux alone, the well $D_i \text{ div } T$ each interval being the record's quanta as the fields' source and no count; (d) *the fall*, for a free packet and for a cloud, $a = \Delta |d\omega_b/dc| \nabla c$, Δ the band's curvature at $k = 0$ and $d\omega_b/dc$ from Rule3's coefficients at the body's pace and rotation; a compact pixel falls by its tail alone, slower, so this line does not hold there. The world file names the body's family, its Nodes with their counts

and the quanta of other families it holds, and nothing else: no momentum, no spin, no emitter, no sign, no seed, no stop, no period; its two levels are the generator's mode file beside the world [1, the four lines of the body].

5.2 The functional and the dilation

Let a body of M quanta have the record ϕ over its region, $\sum \phi_i^2 = 1$, its counts the form $c_i = M\phi_i^2$, and let every held row r its sources be a hollow ($\sigma_r = +1$) or a hill ($\sigma_r = -1$) with the static kernel $\mathbf{G}_r$ of its line, $(\Delta^2 - \kappa_r^2) a = -3 \text{ den } \sigma / \text{num}$ ($\kappa = 0$ for the massless row), and the divisor E_r , so that the well at a Node is $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$. To first order in Φ/Γ the well line raises the one-Node rotation $2 \cos \omega$ by $4g\Phi/\Gamma$, $g = (\text{den} - \text{num})/\text{den}$, and there the fixed point of the binding row is a critical point on the sphere of

$$F[\phi] = \frac{\text{num}}{3 \text{ den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2, \quad (10)$$

the band's pressure less the wells' energy, each well counted once for the record being its own source (Hartree's form; THEOREM: the record is the top mode of Rule3's symmetric form at the paces, $\mathbf{M}(\Phi) = \mathbf{M}_0 + (4g/\Gamma) \text{diag}(\Phi)$ at first order, and $dF/d\phi = 2\lambda\phi$ on the sphere is the fixed point's own equation with $2 \cos \omega_b = 2 \cos \omega_0 - \lambda$). Let $\phi_R(x) = R^{-d/2} \phi(x/R)$ be the body dilated, d the GameBoard's dimension; the pressure is K/R^2 and each row's well energy W_r/R^{s_r} , s_r the row's scaling power at the body's size: $s_r = d$ beyond the row's reach (a contact), $s_r = d - 2$ within the reach or for the massless row (Poisson's $3/(4\pi r)$ in a box) [1, the functional and the dilation].

Theorem 5 (the stable body). *At a fixed point of the binding row of width R the virial $2K = \sum_r \sigma_r s_r W_r$ holds, and the second variation of F along the dilation is $R^2 F''(R) = \sum_r \sigma_r s_r (2 - s_r) W_r$. A body is stable only where this sum is positive; it is a minimum of F where further the Hessian on the modes orthogonal to the dilation and the translations is positive.*

Proof. $F(R) = K/R^2 - \sum_r \sigma_r W_r R^{-s_r}$; $F'(1) = 0$ is $-2K + \sum_r \sigma_r s_r W_r = 0$; $F''(1) = 6K - \sum_r \sigma_r s_r (s_r + 1) W_r = \sum_r \sigma_r (3s_r - s_r^2 - s_r) W_r$: Derrick's scaling, Pohozaev's identity on the GameBoard [1, the functional and the dilation]. $\square$

The signs: $s(2 - s)$ is $+1$ at $s = 1$, 0 at $s = 2$, -3 at $s = 3$: a hollow holds the body where its well falls slower than the pressure and breaks it where it falls faster, and a hill does the opposite. From them: (a) one hollow beyond its reach binds every mass on a chain, is flat along the dilation on a plane, and in a box has one critical point, a maximum along the dilation, a saddle for every mass: the body disperses beyond it and collapses within it; (b) a massless row holds the body: Poisson's well in a box has $s = 1$, $F = a/R^2 - c/R$ has the one critical point $R = 2a/c$, a minimum for every mass (Pekar's body), its width falling as E_g/M ; beside it a hollow beyond its reach adds a barrier, $F = a/R^2 - b/R^3 - c/R$ on a Gaussian body, the critical points $R = (a \pm \sqrt{a^2 - 3bc})/c$, the wide one the minimum and the narrow one the saddle, none where $3bc > a^2$, the collapse; the largest mass with a minimum is $M_{\max} = a\sqrt{E_b E_g}/\sqrt{3b_1 c_1}$ (COMPUTED, the script beside the paper: on the matter pair [4000, 6000] with a binding row of reach 2.02 Links at $\Gamma = 6,000$, $M_{\max} = 4,450\sqrt{E_b E_g}$); (c) the horizon ends a collapse: beyond first order the well line saturates and the Link's pace weakens the pressure, both vanishing together at the pace's floor, so the collapse ends where a Link's pace reaches 0, a frozen cluster of horizon Nodes and no body of ordinary matter; (d) the hill, a hypothesis under its own name, *the hill holds the body*, and (e) *the cooling*, the hypothesis that a body above its minimum sheds its excess as quanta of its own family and radiates light while it breathes, a cold body being one that has finished shedding. What is missing for the theorem to

close: the Hessian beyond the dilation for a set of rows, the functional beyond first order, and the finite GameBoard [1, the functional and the dilation].

The surplus leaves; the mass defect. At a Node of a body's record the local rotation is read from its three levels, $2 \cos \omega_i = (\text{next}_i + \text{before}_i) / \text{now}_i$, against the local band's top; at a fixed point every Node is bound, and a cloud not at its fixed point has a free part, the Nodes whose rotation lies in the band, which leaves at its band's group velocity by Rule3 as it stands, the count's line carrying its quanta with it. The bound surplus does not leave by itself, since a bound mode is stationary and Rule3 reversible (COMPUTED, the script beside the paper: a cloud twice too wide in a static well breathes about its minimum for 6,000 intervals with no settling), so evaporation sheds the free part alone. The mass defect is the form's fall at a kept count (Section 2.6): a standing mode in a well deepening from 300 to 600 at $\Gamma = 6,000$ loses 4.0 percent of its form with its count kept, so a body that binds is lighter in every field it sources by its binding and no lighter in quanta; gravity's source is the well and the count is the baryon number [1, the surplus leaves; the mass defect is the form's fall at a kept count].

5.3 The two branches: the cloud and the compact pixel

The binding row's fixed point has two branches at every count above the edge, and the body of nature is the wide one (DERIVED from the fixed point, computed by the advisor; the reading the owner's decision) [1, the two branches][3, the body of nature is the cloud].

The cloud, the body of nature. The wide branch, the continuum's Pekar body: the count spread over many Nodes, the radius about $2\Gamma/C$ Links for a count C . Its three masses, the well it sources (the sum over its Nodes of $D_i \text{ div } T$, gravity's source), the count it resists with (the inertia, the momentum reading $n = 3Q j_a \text{ div } T$) and the weight it falls with (the momentum identity's force term under a content gradient), agree to the order $(C/\Gamma)^2$, and it falls as a free quantum falls. Along its branch on an open 41-cube at $\Gamma = 6,000$, matter [4000, 6000], $T = 2^{15}$, under the massless row alone (the tension left out as of the order $(C/\Gamma)^2$ for a wide body), in the advisor's unit, the vacuum's share, which for a cloud is the engine's count to a percent (Section 3.2) [1, the rows against nature, (u)]:

Total count	$2 \cos \omega_b$ (top 1.3333)	rms radius	Centre Node	Well / count	Inertia / count	Inertia / well	Fall / free
4,000	1.3509	8.7	2.6	0.985	1.061	1.077	0.92
3,000	1.3437	10.3	1.0	0.991	1.037	1.047	0.95
2,000	1.3383	11.5	0.4	0.995	1.022	1.027	0.97
1,500	1.3359	12.0	0.3	0.996	1.016	1.020	0.98

The two lighter clouds lean on the box's faces, so their true radii are larger and their numbers nearer 1 than the table shows. So Galileo and Eötvös hold in the law for wide bodies to $(C/\Gamma)^2$, the better the larger Γ is against the body's count. In the integer engine a cloud is sparse: at 1,500 in all there are 0.3 quanta at the densest Node, so the lay writes a quantum at a few hundred Nodes and none elsewhere, and the count's line carries those quanta through a wave shared by ten thousand Nodes.

The compact pixel, collapsed matter. The narrow branch: one Node and its tail, a standing record above the top of its family's band in the wells its own form lays, its count the sum over its Nodes, at one Node near the horizon and over its tail below it; a cluster is pixels within tail reach that click at one another. Its three masses part by the binding, since the gradient reaches its quanta only through its tail, where its own well has taken most of the Link pace $((1 - 2x_0)^2 = 0.14$ at 2,000). Along the narrow branch under the massless row with the tension (the counts in the

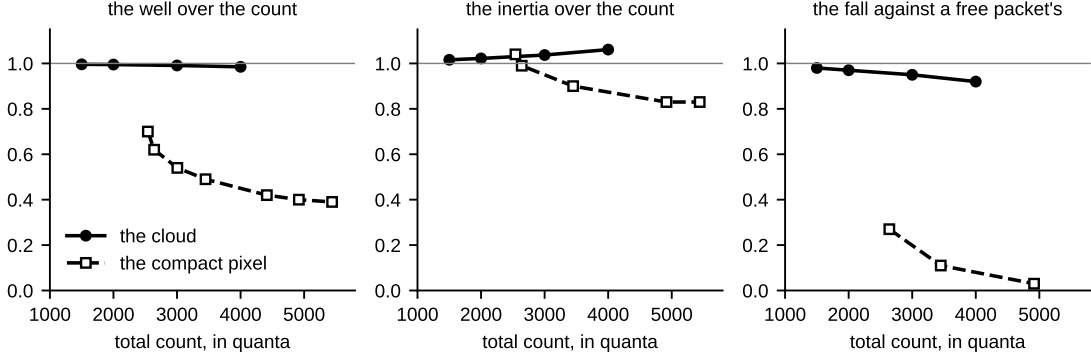


Fig. 5 The three masses of a body along its two branches, from the documents' rows (DERIVED from the fixed point, a reading of the algebra and no click; the counts in the advisor's unit): the well over the count, the inertia over the count and the fall against a free packet's, for the cloud (the wide branch, filled marks) against its total count, and for the compact pixel (the narrow branch, open marks) against its total count. For the cloud the three go to one and the fall to a free quantum's the wider it is; for the compact pixel they part by the binding. Drawn from the rows of the law's document; no run.

advisor's unit; the engine's count about three times them, 3.7 at the Node and 3.1 in all at 2,000) [1, the rows against nature, (u)]:

Count at the Node	Total count	The well	Well / count	The inertia	Inertia / count	Fall / free
1,650 (the edge)	2,541	1,769	0.70	2,642	1.04	
2,000	2,640	1,637	0.62	2,621	0.99	0.27
2,500	3,005	1,636	0.54			
3,000	3,449	1,692	0.49	3,105	0.90	0.11
4,000	4,413	1,860	0.42			
4,500	4,917	1,967	0.40	4,069	0.83	0.03
5,000 (near the horizon)	5,440	2,114	0.39	4,517	0.83	

A free quantum of matter at rest has a well equal to its count (a plane wave at $k = 0$ carries per Node the count $b^2(1 - (\text{num}/\text{den}) \cos \omega_0)/T$ and the well $b^2 \sin^2 \omega_0/T$, equal at $\cos \omega_0 = \text{num}/\text{den}$), and a compact pixel's well is under its count everywhere in the window, 0.70 at the edge to 0.39 at the horizon: the binding lightens, the more so the deeper. Its inertia lies near its count and not its well, and its fall is suppressed, none at the horizon: a compact body sources gravity with its well, resists with its count and falls with its tail, the three masses parting by the binding. The horizon is a horizon, the pixel at the horizon a collapsed body and no proton, and gravity through the pace is not a metric for a compact body: a metric moves a body and its binding together, here the gradient moves the record and leaves its wells to follow, and the record is sheltered in them. That is the law's word on collapsed matter and no finding against nature, since nature's bodies are wide (Fig. 5). The well has a minimum along the branch, 1,636 at about 3,000 in all, so accretion pays all the way to the horizon, where it ends. Between the two branches stands the saddle, where the total count falls with the binding (the Vakhitov and Kolokolov reading, $dC/d\lambda < 0$ with $\lambda = 2 \cos \omega_b$) [1, the three masses of a body; the two branches].

The electron and the proton. The electron is one quantum of the matter family with the sense -1 , a message and no pixel, stable by the integer (one quantum does not disperse into fractions), its charge the proton's exactly since the sense is a sign, the positron the same quantum in the sense

+1; it needs no band of its own. The proton is a cloud of 1,836 quanta, so $m_p/m_e = 1,836$ is the count's ratio directly, and it fixes no Γ : Γ is free and large against 1,836, as nature's Link is small against a proton. And 1,836 is not derived: the law has one scale, Γ , and pure numbers of order one (the pair $2/3$, π , the lattice Green's function), so every count it derives is a fraction of Γ or a small number, and a ratio of 1,836 between two bodies needs Γ or a large pure number the law lacks; the roads tried (the collapse count, about 0.5Γ ; the lightest cloud that binds at all, about $\sqrt{8.4\Gamma}$; their ratio; the coincidence $6\pi^5 = 1,836.1$) give none. So the proton's count is an input, an integer the world file declares as every body's count is, and the law fixes the electron's, one (ASSUMPTION, the owner's decision on the advisor's computation) [1, the electron needs no band of its own; the rows against nature, (t)].

The lightest body, and the pair tied to Γ . Under the two rows a cloud stands only within the range R : under [600, 601], $R = 10$, the cloud of 4,000 stands ($2 \cos \omega_b = 1.3367$, the rms radius 10.4, its well 0.995 and its inertia 1.02 of its count, its fall 0.975) while the clouds of 1,836 and 1,000 do not (their Poisson radii 12.2 and 12.7 Links, over the range); the binding holder holds no cloud lighter than about 1.5 times $2\Gamma/R$, so there is a lightest body, the lightest cloud is the lightest baryon, the binding holder's own quanta are the mesons, and the electron is never a body. The cloud of 1,836 is not bound under $R = 10$ nor under $R = 15$ ([1350, 1351], 1.33320, under the top by a hair) and is bound under [2400, 2401], $R = 20$ ($2 \cos \omega_b = 1.33394$, the rms radius 11.9, its well 0.997 and its inertia 1.011 of its count, its fall 0.988 of a free packet's), so the file's pair is [2400, 2401] and R is tied to Γ by the proton, one number fewer in the file (COMPUTED) [1, the rows against nature, (w)][3, the body of nature is the cloud].

The charged cloud, and the proton against the neutron. The charge is the booking W , read into the pace at the weight 1 as gravity's well is, so a cloud whose every quantum carried charge would have a hill of the sign holder equal to its gravity well and would not bind; the proton stands as a cloud of 1,836 quanta with a total W of one unit, its imaginary part a thousandth of its real one, one mode of one sense whose rotation is a thousandth elliptical. Derived beside it, a finding against nature: the charged cloud is bound a little less than the neutral one of the same count, by its own hill, so it is heavier by about 10^{-5} of its mass, where nature's neutron is the heavier, by 1.4×10^{-3} , opposite in sign and smaller [1, the two branches; the rows against nature, (z)].

5.4 The generator

The generator builds every world from zero, declares nothing and holds no float: the fixed point of the binding row by the law's own iteration, the mode under the wells its form lays and the wells from the mode, until the levels repeat; the two levels scaled together so that the weighted share the record lays carries the body's count; the held rows at their rest, the fixed point of the division act with its remainder, computed once before the first interval and never in it; a body laid with a sense carrying its second level pair, the record a quarter period on; a message laid by the file's keys alone, the wave under its envelope with its level one interval earlier, every cosine by Rule3's rotation act. The branch is selected by the start's width alone: from a wide start the iteration reaches the cloud, from a narrow start the compact pixel, two fixed points at one total, and the iteration goes to the one its start is nearer to, never across. The amplitude unit A is derived from the integer width as the largest amplitude at which Rule3's total stays inside it, never written [1, the generator][3, the generator is Rule3].

6 The forms of nature the line arrives at

Each form below is arrived at from the line before any run and is a prediction of the law, never a result; the form of nature beside it is on the comparison side and is never an input. The coefficients of Eq. (1) were written for the Node: a held level lowers the clock's pace once and the Link's pace once more, with no metric, no boost and no potential in them; what follows is what those coefficients do to a packet, read by the geometric optics of its band. $U_b = c_b/\Gamma$ is the potential at the point named, the clock's shift per level [1, the rows against nature].

- (a) *The redshift* (DERIVED). Two clocks of one family at the levels c_1 and c_2 have their tick intervals in the ratio of the root of $(1 - 2U_1 + 2U_1^2)/(1 - 2U_2 + 2U_2^2)$, $1 - (U_1 - U_2)$ to first order, for a band whose rotation follows the clock's pace; the band's own slope: at the content x a record of the pair [num, den] at rest rotates at $2 \cos \omega = 2 - 2((\text{den} - \text{num})/\text{den})((1 - x)^2 + x^2)$, so for matter [4000, 6000] $d\omega/\omega = -1.06x$ to first order and the gravitational redshift is the clock's share, $z = 1.06x$; light born by a breathing body in the well leaves with the body's rotation and keeps it in flight (a static content changes no frequency), so a far detector reads the emitter's slow clock. Beside it: Einstein's gravitational redshift.
- (b) *The bending* (COMPUTED). At a Node of content x light's index is $n = 1/(1 - 2x)$, $n - 1 = 2x$, the Link's pace reading the content twice (Section 2.5); far from a body its field is the massless row's rest, $L(r) = 3\sigma_{\text{tot}}/(4\pi r)$ beyond a few Links, σ_{tot} the body's wells per interval, so $n - 1 = 3\sigma_{\text{tot}}/(2\pi\Gamma r)$ and a beam past the body at the closest distance b turns by $3\sigma_{\text{tot}}/(\pi\Gamma b)$ radians, $4U_b$, the centroid at a receiver L Links beyond shifted by $4U_b L$ toward the body. The ratio that is nature's: the bending is $2x$ and the redshift $1.06x$, so light is bent by 1.9 times a matter clock's share (exactly 2 against the clock's own share x , the 1.06 the band's slope at [4000, 6000]), against nature's 2, with no chosen number in it. By the geometric optics of the band from the coefficients themselves (the script beside the paper, both forms of the self coefficient): the deflection is 2.02 times Newton's $2U_1/b$ with the clock to the first order or the second, Einstein's $4U_1/b$; with the Link's pace lowered once it would be 1.01 times Newton's. Beside it: Einstein's $4GM/(c^2 b)$.
- (c) *The delay* (DERIVED). A round trip past the body is delayed by $4U_b$ times the path's length within the well's reach. Beside it: Shapiro's delay.
- (d) *The perihelion* (COMPUTED). An orbiting emitter at the semi-axis a and the eccentricity e advances $6\pi U_p$ per orbit with U_p the potential at $a(1 - e^2)$: by the same geometric optics the matter pair's periapsis advances $7\pi U_1/(a(1 - e^2))$, 1.17 times Einstein's, with the clock to the first order and $6\pi U_1/(a(1 - e^2))$, 1.00 times Einstein's, with the clock's second order, which is the line (Section 2.5); the bending stays 4 under both. Beside it: Einstein's $6\pi GM/(c^2 a(1 - e^2))$.
- (e) *The moving clock* (DERIVED). A body at the velocity $v = n/W$ (its phase k per Link an exact pair of the file) has its tick lengthened by the dispersion of its own bound band: the tick's factor is $f = \Omega(k)/\omega_0$ with $\Omega(k) = \omega(k) - k\omega'(k)$ the rotation at the moving centre, the phase rate less k times the group velocity, and $\omega(k) = \omega_0 + \Delta(1 - \cos k)$ the body's bound band, ω_0 its rest rotation and Δ its packet's top velocity, both the generator's from the file's counts and pair, no key; to the second order $f = \sqrt{1 - v^2/c_b^2}$ with $c_b^2 = \Delta\omega_0$, Lorentz's factor at the body's own light speed; read as the mean of the front and back click periods $P_v(1 \mp v/u)$ at two resting detectors, u the light's group velocity, $P_{\text{rest}}/P_v = f$. Beside it: Lorentz's factor, at the body's own light speed and not the lattice's, since no boost is among the 48. Lorentz is arrived at as the symmetry of the clicks Outside, not as a symmetry of the lattice [3, the click].
- (f) *The charge* (DERIVED). Like senses a hill, unlike a hollow; a reader of the sense q reads $-q$ times the holder's level in its pace. The coupling goes as the sign: the read by sign enters the pace as

q times a level, never as Q/M , so an electron and a proton accelerate alike in one field, and the response in a gradient is the mode's own and not the coupling's. A current's read (the magnetic force, frame dragging) is a hypothesis under its own name and no line of the law. Beside it: Coulomb's force.

- (g) *De Broglie's relation* (DERIVED). A free record of a matter family travels only where its band holds its rotation, $\text{num}/(3 \text{ den}) \leq \cos \omega_b \leq \text{num}/\text{den}$; its wave number is the band's, $\cos \omega_b = (\text{num}/(3 \text{ den}))(2 + \cos k)$, its group velocity $v = (\text{num}/(3 \text{ den})) \sin k / \sin \omega_b$, and to the second order $k = \omega_b v / c_m^2$ with $c_m^2 = \omega_0 / (3 \tan \omega_0)$, the family's own light speed. Beside it: $\lambda = h/p$ with the rest rotation as the mass.
- (h) *The fall* (DERIVED). In a pace gradient a free packet or a cloud accelerates at $a = \Delta |d\omega_0/dc| (c_+ - c_-)/D$ toward the well, c_+ and c_- the read level at its two faces D Links apart, so bodies with the same $\Delta \times d\omega_0/dc$ fall alike, and a cloud falls as a free quantum falls to $(C/\Gamma)^2$ (Section 5.3); a compact pixel falls by its tail alone, slower. Beside it: Galileo's fall and the equivalence principle, which hold in the law for the bodies of nature and not for collapsed matter.
- (i) *The pull and the orbit* (DERIVED). The massless row rests at Poisson's with the sources and at Laplace's outside a body, $c(r) = 3s/(4\pi r)$ falling to the faces' 0, so the acceleration falls as $1/r^2$ and a circular orbit's v^2 as $1/r$; under the two rows a reader within R reads the binding holder's screened field instead, and a row that reads $1/r$ beyond R takes $E_g = 1$ in its own world. Beside it: Newton's pull and Kepler's fall-off; a flat $v(r)$ would name a missing law.
- (j) *The medium* (DERIVED). Inside a body of pace p light of wave number k is slowed to $v_{\text{in}} = (p/\Gamma)^2 \sin k_{\text{in}} / (3 \sin \omega)$ with $1 - \cos k_{\text{in}} = (\Gamma/p)^2 (1 - \cos k)$, the index v_g/v_{in} , and is reflected whole where $p < \Gamma \sin(k/2)$; through a window moving at w the wave number is matched at the moving face, $\omega - kw = \omega_{\text{in}} - k_{\text{in}}w$, and the speed inside is the band's group velocity at that k_{in} . Beside it: the refractive index and the mirror; Fresnel's drag $w(1 - v_{\text{in}}^2/v_g^2)$ is nature's number beside it and not derived from it.
- (k) *The round trip and Sagnac* (DERIVED). A radiating body receding at v from a mirror at rest reads its returned clicks every $P(u+v)/(u-v)$ intervals for a write every P , u the group velocity at the wave number that solves $\Omega(k) + kv = \omega_b$; a radiating body moving at v along a ring of N Links reads its two arms $2Nv/(u^2 - v^2)$ apart, $4A\omega/u^2$ for the ring's area A and angular speed ω to first order. Beside them: the two-way Doppler ratio and Sagnac's $4A\omega/c^2$, with u in place of c .
- (l) *A free record keeps its wave number* (THEOREM). Rule3 is translation-invariant, so a free record keeps its wave number and rotation exactly and a radiating body's period is read unchanged at every distance, the record's rows widening by the dispersion while its count does not move, the rounding's walk at most one unit of level per Node per interval. Beside it: Hubble's drift, which the law does not give from a fixed Γ ; the law's own line for an expansion is the hypothesis *Gamma is not constant* (Section 8.4).

The numbers of one body. For the 2,000 pixel at $\Gamma = 6,000$ (COMPUTED, the advisor's derivation): under Poisson's rest, the bending $0.26/b$ radians, 3.0 degrees at $b = 5$ Links and 1.5 at 10, a $\lambda = 8$ packet passing at $b = 6$ turning by 0.043 radians, a transverse shift of 1.3 Links over 30 Links of path; a matter clock five Links from the pixel ($x = 0.013$) running 1.4 percent slow against a far one. Under the two rows the massless row's field is over E_g and a reader within R reads the binding holder's screened field instead [1, the rows against nature, (a) and (b)]:

Distance from the pixel	The field, in levels		The clock's share z		The bending at that impact parameter	
	[2400, 2401]	Poisson	[2400, 2401]	Poisson	[2400, 2401]	Poisson
1 Link	373	386				
2 Links	189	205				
3 Links	117	133			4.9 deg	5.0 deg
5 Links	62	79	1.09 %	1.4 %	2.8 deg	3.0 deg
10 Links	24	40			1.2 deg	1.5 deg

The bending stays 1.9 times the clock's share under both; the lattice computation of it to three digits, a pixel of 2,000 and a packet passing at five Links, the count's line's deflection against the clock's share at the same Node, is the first proof-grade number the law offers (Section 8): 2.00 within the rounding is the direction confirmed, 1.9 sharp a lattice correction to state [1, the rows against nature, (b) and (v)].

How the forms are reached, not assumed. Nothing above was put into the coefficients: no metric, no Lorentz transformation, no potential and no force. The one thing the law says is that a held level slows a Node's clock once and its Links twice, and that a record's rotation follows its band. The bending's 4 and not 2 is the Link's second factor; the perihelion's 6π and not 7π is the clock's second order, the composition of the slowing unit by unit; Lorentz's factor is the Legendre transform of a bound band read at its moving centre. Where the law's number differs from nature's form at the declared numbers, the difference is the law's own, read blind: the bending's 1.9 against 2 (the band's slope at the matter pair), Sagnac's u against c , Fresnel's drag not derived.

7 The two slits

The first experiment run on the engine, and the first row of this paper with a result: a world declared in the law's files, a blind row written before the run from the law's own lines, the run on a named commit of the engine, and the detector's clicks beside the blind row. Every number here is the law's document's, row (g) of the rows against nature, and is cited so [1, the rows against nature, (g)][3, the two slits]. The rule of the row: an experiment checks only clicks; its operator, its tools, its conservation and its initial state are proved in the algebra and checked at load, never by a run; a run before its blind number is a first look and says so [3, an experiment checks only clicks; algebra first].

7.1 The world

The emitter is a laid message, a packet of the light family (the charge, the sign holder's own real record), no body: a wavelength of 8 Links along $+x$, $k = \pi/4$, $\cos \omega = 0.9024$, the period 14.1 intervals, the group velocity 0.547 Links per interval (light's speed at long wavelength is $1/\sqrt{3}$); its levels laid at the first act as $\text{now}_i = b e_i \cos(kx_i)$ and $\text{before}_i = b e_i \cos(kx_i + \omega)$, the wave one interval earlier, with $b = 1,328$ so that the lay carries 10.00 quanta per Node of the plane wave, under a raised cosine along x over $x = 4$ to 16 centred at $x = 10$, flat across $y = 4$ to 44 with edges of two Nodes; a real packet, neutral. The wall is not matter (a mirroring row of pixels is past the horizon by its summed fields) but the board's own face rule declared inside the board: the inner face at $x = 20$ with two gaps of three Nodes, $y = 17$ to 19 and 29 to 31, the slits' centres 12 Links apart, a declaration of the file like the wraps and the detector Nodes, no number in the engine. The board is flat, $64 \times 48 \times 1$, the third axis folded, every face open. The screen is the 48 detector Nodes of the column $x = 44$, one per y , $L = 24$ Links from the wall. The run is 130 intervals; the packet's centre reaches the wall at about interval 18 and the screen at about 62; the screen is read in the window

of intervals 45 to 120, before the far face's return. The universe file is the rule's own, $\Gamma = 6,000$, $T = 2^{15}$, gravity at the divisor 1,000 and the binding holder [2400, 2401] (Section 4.2). Three worlds: the packet as above; the dilute packet, $b = 420$, one quantum per Node; and the which-way world, the same packet with a detector Node in the lower gap, [20, 18, 0] [1, the rows against nature, (g)].

The folded board is the experiment exactly and no special case: for a record uniform along z the folded axis returns the Node itself through both z Ports, so light's band on the flat board, $2 \cos \omega = (2/3)(\cos k_x + \cos k_y + 1)$, is the three-dimensional band at $k_z = 0$, and the screened sums are the three-dimensional ones summed along z ; a thick folded board with the same packet gives the same fringes with the counts multiplied by the thickness, and a thick open board is a second world (DERIVED) [1, the rows against nature, (g)].

7.2 The blind row

Written before the run, from the law's own lines (COMPUTED): light's plain line on the very board in real arithmetic, the count through each screen Node the integral of the current through the screen column over the window, $\sum F \operatorname{div} W_c$, the detector's rises summing to it within the rounding. The formula: the fringe spacing $\lambda L/d = 8 \times 24/12 = 16$ Links, maxima at $y = 24, 8$ and 40 , minima at $y = 16$ and 32 , under the three-Node gap's envelope. The numbers, DETECTOR: through the screen in all 273 quanta (15 percent of the packet, the gaps' share less the wall's return); at $y = 24$ 13; at $y = 16$ and 32 1.3; at $y = 8$ and 40 8 and 9; across $y = 0$ to 47 the row of Table 1, the pattern on $y = 4$ to 43 (the outer four Nodes carry the y faces' reflections). At most 2.3 quanta stand on a screen Node at any interval, so the rises come one at a time, about 13 at the centre over the window and 1 or 2 at a minimum; the visibility of the rises about 0.85, the minima 0 to 2, the rounding's share. For the dilute packet: only the Nodes whose flux crosses $W_c/2$ report, the maxima first, the pattern thresholded and not sampled. For the which-way world: a detector Node in one gap changes nothing of the row, since nothing absorbs [1, the rows against nature, (g)].

7.3 The result

The run of 2026-09-30 on the two rows' engine, every held row stepped: the back-in-time gate says MATCH over 130 intervals on each of the three worlds, and the three runs are LAWFUL over their 130 intervals. DETECTOR, the screen's rises over the window 45 to 120 (Table 1 and Fig. 6): through the screen 258 quanta in all against the blind 273, every rise between the intervals 61 and 99; at $y = 24$ 18 against 13; at $y = 16$ and at $y = 32$ 0 against 1.3 each; at $y = 8$ 5 with 7 on either side against 8, and at $y = 40$ 10 against 9; the spacing 16 as the formula says; the visibility of the central maximum against the two minima $(18 - 0)/(18 + 0) = 1$ against the blind 0.85. The central maximum and the two minima stand where the formula puts them, the maximum higher and the minima emptier than the blind row, the side fringes under the rounding's ripple, the rises one at a time; the reader's own local extrema of the row fall at $y = 4, 7, 9, 12, 15, 19, 24, 36, 40$ and 43 and at $y = 8, 11, 20, 38$ and 41 , the rounding's walk between the fringes. Against nature: the fringes are there in the aggregate count, at the spacing 16 Links, from a run of the local click alone. About one quantum in seven passes the gaps, riding the two fans, and the wall's reflection goes back toward the source [1, the rows against nature, (g)].

7.4 The two findings against nature, and the third reading

- (i) *The dilute world.* The packet of one quantum per Node (168 quanta laid) gave no rise on the screen over the whole run, not at the maxima either (its 53 clicks are the open faces' layer's, from the interval 53): the pattern thresholded to nothing, as the blind row allowed and further.

Table 1 The two slits, DETECTOR: the count through each Node of the screen over the window, blind (written before the run) and measured (the rises), across $y = 0$ to 47; the pattern is read on $y = 4$ to 43. From the law's document, row (g).

y	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
blind	2	6	7	6	3	3	6	8	8	6	6	6	6	4	3	2	1	2	2	4	7	9	11	13
measured	0	4	10	4	6	0	0	7	5	7	6	3	7	1	1	2	0	0	6	7	6	8	12	17

y	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
blind	13	12	11	9	7	4	3	2	1	2	3	5	5	6	6	8	9	6	3	4	7	8	7	3
measured	18	17	12	8	7	6	6	0	0	2	2	2	5	4	2	3	10	1	3	4	9	10	6	2

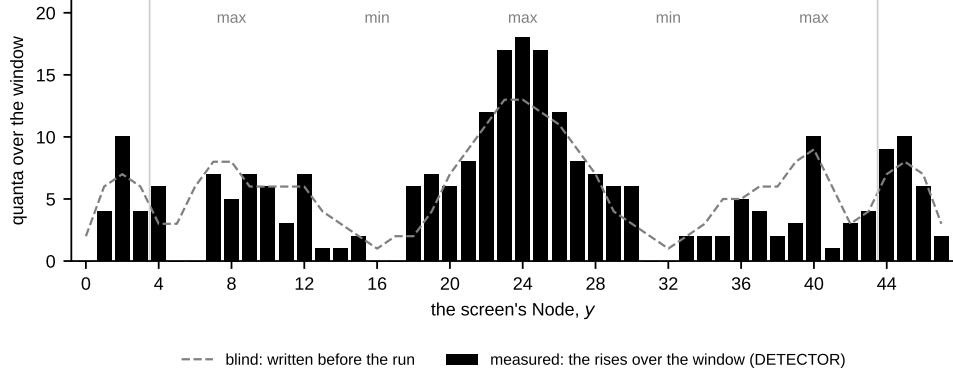


Fig. 6 The two slits: the screen's rises over the window (the bars, DETECTOR, the measurement) beside the blind row written before the run (the dashed curve), across the screen's 48 Nodes; the pattern is read on $y = 4$ to 43, and the outer four Nodes on either side carry the y faces' reflections. The central maximum at $y = 24$ and the minima at $y = 16$ and 32 stand where the formula $\lambda L/d = 16$ puts them. Drawn from the rows of the law's document; no run of this paper's.

The silence is the cold board's: every remainder was 0 at the start, and a cold detector rises only on a whole quantum (Section 3.7). So the dilution series is run on a warm board, a bright packet first, leaving its remainders at the screen and everywhere it passed, then the dilute packet on the same board, or several dilute packets in a row, each warming the screen for the next; its blind row, written here before the run: about one rise per dilute quantum in the mean, distributed along the screen as the fringes, the pattern built quantum by quantum as in nature's single-quantum experiments, and on the cold board none, as run. No single-photon statistics are claimed from the run so far; the run to come is the law's answer.

- (ii) *The which-way world.* The detector Node in the lower gap reports 122 quanta entering it from the interval 12 (48 within the window), and the screen's row is the same 258 rises, Node for Node, as without it: a reporting detector in one gap changes nothing of the row. That is the law's word that a bare detector changes nothing, not a match with nature, whose which-way loss of the fringes needs the absorber of the absorption look (Section 7.5).
- (iii) *The centre's excess, read as the light's self-focusing.* The minima and the total are within the rounding the blind row itself named; the centre is 38 percent over and the side fringes about 20 percent under (the central fringe's nine Nodes 105 against 92, the left fringe's nine 41 against 52), the pattern a little more concentrated than the blind row. The advisor's reading: the blind row was the plain line with no wells, and the run had the packet's own well read back by the light through the binding holder at the divisor 1 (ten quanta per Node lay a well of about ten levels per Node, its screened field some tens), so the light bends toward its own two fans,

a self-focusing of light by its own well at the coupling 1, which nature does not show at any intensity: a finding beside the fringes and not the rounding's, to be checked by the advisor's real-arithmetic run with the packet's well in it (DERIVED, a reading and not yet a computation) [1, the rows against nature, (g)].

Two caveats the row carries as its own. The runs stood with the vacuum's row's tensions 0 at the start (the start sourced the time parts alone); with the tension's source corrected in sign the packet's own stress deepens its content by a few levels and the fringes move by less than the rounding, so the run did not wait for it. And the runs stood before the vacuum content entered the file; under c_{vac} light's long-wave speed is 2 percent under the vacuum's, uniformly, the fringes' spacing $\lambda L/d$ unchanged and only the arrival shifted by about one interval, which the same reading covers [1, the rows against nature, (g)][3, the two slits].

What the look shows and does not claim. The bars are the measurement, the only numbers compared with nature; the glow and the dots on the board are a GameBoard reading, labelled so; the wall is a declaration of the board's shape; the dashed blind curve stays behind the bars, labelled blind, never on the board; no single-photon statistics are claimed; the dilute world's no rise is the cold board's silence and the which-way world's identical row the law's word that a bare detector changes nothing, no match with nature [3, the two slits].

7.5 The looks to come, blind

Three looks stand in the law's document with their worlds and their blind rows written before any run, and none has run as this chapter is written; the paper carries them as the documents hold them and marks each not yet run. Each is a GameBoard reading, a diagnostic and no click, but for what its detectors report [1, the rows against nature, (s), (w), (x), (y)].

The pixel look, collapsed matter. One pixel of 2,000 at the Node on the rule's own universe, on the open 41-cube, the generator laying the standing mode with its tail and the start sourced by the wells the lay writes. The blind row, final, under the two rows, the polarisation row stepped, the tension read with its source $-G$ and the vacuum content $c_{\text{vac}} = 60$, every content carrying $x_{\text{vac}} = 0.01$ and the vacuum's band top 1.3465 in place of 1.3333: $2 \cos \omega_b = 1.4422$ against that top (the binding 0.096 over it), the period 8.21 intervals, the content at the Node $x_0 = 0.2835$, the contact 608 and the field 1,033 at the Node, the total 3,256 in the advisor's unit (the engine's count by the generator's conversion at the Node's paces, about three times these at this depth), the well over the count 0.701 and the inertia over the count 1.188 in the advisor's unit, the rms radius 1.1 Links, the fall 0.354 of a free packet's, the pace far from the body $\Gamma - 120$ and at the Node $p_a/\Gamma = 0.411$; at 3,000 at the Node $2 \cos \omega_b = 1.5282$, the period 8.96, $x_0 = 0.3300$, the contact 752 and the field 1,168, the total 3,640, the well 0.552 and the inertia 1.129 of the count, the rms radius 0.6, the fall 0.154; the tail's tension deepens the well, so the body is a little tighter and its fall a little more suppressed than under the earlier sign. Over a run of at least three periods the form returns within the mode's level at the Node, the field within one unit after the transient, the count within its rounding walk of 0.058 quanta per unit; the back-in-time gate before the look; the look with the vacuum content in the file (COMPUTED, not yet run) [1, the rows against nature, (s)].

The cloud look, the body of nature. One cloud of about 4,000 quanta on an open board of 41 Nodes or more at $\Gamma = 6,000$, laid real, the generator laying the wide branch's fixed point from a start of the width $2\Gamma/C$: it stands at $2 \cos \omega_b$ above the band's top with its count kept, a quantum at about 500 Nodes and none moving in the mean, the quanta scattered in one glow shared by ten

thousand Nodes; under [2400, 2401] with the recomputed numbers, the cloud a little less bound, its tail cut at R (COMPUTED, not yet run) [1, the rows against nature, (w)].

The fall look, the two branches in one gradient. The board a tube, y and z wrapped at 41 each, x open and 81 long; the source a compact pixel at $x = 60$ whose own fall is 0.03 of a free packet's, so it stays; the falling body at the tube's middle, where the pushes of its own two images cancel; three worlds with the same tube and source, one falling body each: a pixel of 2,000, a cloud of 4,000 and a free packet (matter at $k = 0$, a Gaussian of rms 6 Links, about 50 quanta). Under the two rows the tube reads the binding holder's tail: the gradient at the body 0.97 levels per Link, the free fall 4.3×10^{-5} Links per interval²; a uniform c_{vac} adds no gradient, and the free packet's fall at x_{vac} is 2 percent under the vacuum's; the blind drifts under the pixel's final row, read as the count's centroid along x , over 300 intervals 1.9 Links for the free packet, 0.68 for the pixel of 2,000 (its factor 0.354) and 1.9 for the cloud of 4,000 (0.99), over 200 intervals 0.85, 0.30 and 0.84, the cloud's mean gradient 1.02 of the axis value; the pixel against the cloud is the row, 0.35 against 0.99, the free packet beside them, and the cloud's 0.92 to 0.99 stands under every version (COMPUTED, not yet run) [1, the rows against nature, (x)].

The absorption look, nature's detector. The owner's two blind rows for the build of the sense: a body of rotation ω_b in a light beam at its resonant frequency absorbs and at another frequency does not, the line of a spectrum, the law's first; and the light count that arrives at the absorber and stays is what its detector Node reports, one photon at one absorber one click. Computed with the read alone, the write not built, on the 2,000 pixel's fixed point in real arithmetic with a plane wave read into the body's content by its sense at the weight 1 (COMPUTED, the assumption of the weight stated): (a) the line is parametric, at twice the rotation, $2\omega_b$ (the period 4.25 intervals), the form times 1.87 at $2\omega_b$ over 100 intervals at the level 60 against 1.002 at ω_b and 1.01 at $\lambda = 8$; (b) the resonant light is off-axis and lattice-scale, on the (1, 1, 0) diagonal with $k = 1.94$ per axis, since a body's rotation is at matter's own scale; (c) photo-ejection, the law's photoelectric effect, with the threshold at the binding: the body sheds record and count as a free wave when $\omega_L > \omega_{\text{top}} - \omega_b = 0.10$ (λ under 36 Links), 0.065 quanta per interval at $\lambda = 8$ and none under the threshold, proportional to the level squared; (d) one quantum per Node of light at $\lambda = 8$ is a level of 420, 7 percent of Γ , so the beam is at the level 60. The world: the 2,000 pixel at rest on the 41-cube with open faces, a plane packet of the light family laid on one side, 100 intervals, three beams in three worlds (the parametric beam at $2\omega_b$, the $\lambda = 8$ beam, a beam under the threshold at $\lambda = 72$). The decisive row: the body's form per interval against the light's count that stays at its Nodes: if the read and the write are one coupling, the body's re-write cancels the beam at its Nodes and the form's gain is bounded by the arriving count, at most 0.45 well units per interval; if the form rises at the read's rate while the light passes undepleted, the law makes form from nothing, a finding of the first order. The blind rows: (i) the parametric beam, the form's rise per interval equal to the light's count staying per interval, at most 0.45; (ii) the $\lambda = 8$ beam, 0.065 quanta of matter per interval leaving the body as a free wave, the body's form within 1 percent; (iii) under the threshold, no matter leaves and the light's count passes whole. The dilution series and the one-photon click come after this row, since they need the write [1, the rows against nature, (y)].

8 The constants, and what is open

8.1 The constants

The causal bound is one Link per interval, and light's own speed at long wavelength is $1/\sqrt{3}$ Links per interval (Section 2.2). The quantum of action is one click, its energy the quantum's norm T of

its family. The quantum of distance is the Link, of time the interval; nothing between two Nodes or two intervals is observed. Newton's constant is not declared: a body whose wells are s quanta per interval makes the massless row's rest $3G(r)s$ at r Links over the row's divisor, $0.7582s$ at the Node and $3s/(4\pi r)$ far away, and the level slows the clock by the potential $\Phi = -c/\Gamma$, so (DERIVED)

$$G = \frac{3}{4\pi\Gamma E_g} \text{ per quantum of well, in Links and intervals,} \quad (11)$$

a reading of Γ , the file's E_g and the lattice, and no number of the law; a world at a smaller Γ has stronger gravity per unit of content, and the ratios between rows carry G once and cancel it. The row's source moves by whole quanta of well and never by less, and the field itself is no click. The vacuum content c_{vac} is the law's cosmological constant, positive as nature's, one more number of the file (Section 4.5) [1, the constants].

The free numbers. Under every divisor 1, G and α are no numbers of the law but readings of Γ and the GameBoard's geometry; under the two rows the file holds gravity's E_g beside Γ , T and the matter pair's m , the binding holder's gapped pair (its range R tied to Γ by the proton) and c_{vac} . Γ is free and large against a proton's count, as nature's Link is small against a proton, and no ratio fixes it yet (item 19 below); α stays a reading; m is the one integer of nature beside them, bounded by the band and the reach [1, no free number but the mass].

T is physical through $\sqrt{T}/\Gamma$. T is not the resolution only, though the rounding's floor stays (2^{19} for a hundredth of a quantum per unit at the pixel): the level of one quantum of light is $\sqrt{T}/\sin\omega$, from $b^2\sin^2\omega/T = 1$, and the pace reads levels, so light's action per quantum on a body's content is $\sqrt{T}/(\Gamma\sin\omega_L)$, 0.07 of Γ for one quantum per Node at $\lambda = 8$ at $T = 2^{15}$ and $\Gamma = 6,000$ (0.03 the floor at the shortest waves, 0.28 at $T = 2^{19}$); matter's well per quantum is 0.76 levels, so at these T and Γ light acts on a pace 550 times more per quantum than matter's gravity. The pair (T, Γ) holds one physical number, $\sqrt{T}/\Gamma$, light's coupling per quantum, and nature's is small; what fixes it is open beside Γ [1, what is open, item 9].

8.2 The constants the law derives with no free number

A constant of nature is derived as a dimensionless number, or as a dimensional one once the units are fixed from nature; five candidates, in the order of their computation, each written before its run and each written as it comes, against nature or not [1, the rows against nature, (v)]:

- (1) *The bending factor of light*, the bending over the clock's share, Einstein's 2 against Newton's 1: 1.9 on the lattice in the first approximation (Section 6, (b)); the lattice computation to three digits is the first proof-grade number, 2.00 within the rounding the direction confirmed (COMPUTED in the first approximation).
- (2) *The strength of gravity against the quantum*, $Gm^2/(\hbar c)$: under every divisor 1 the coupling of one quantum to another through the content holder is of order one, against nature's 1.75×10^{-45} for the electron and 5.9×10^{-39} for the proton, and the same number read between two charged clouds is the hierarchy (Section 4.4); decided as the two rows with the caveat, the hierarchy the file's one number, E_g about 3×10^{42} for nature, and the charged cloud look reads it as $E_g/1,836^2$; the finding of scale beside it (COMPUTED under every divisor 1; ASSUMPTION under the two rows).
- (3) *The fine-structure constant*, α : the sign holder's well of one quantum with the sense against the content holder's well of one quantum, both at the divisor 1, the electric against the gravitational pull between two electrons, nature's 4.2×10^{42} , and α itself as light's coupling to the charge, to compute, with its distance from $1/137.036$ (open).

- (4) *The mass ratio $m_p/m_e = 1,836$* is the count's ratio of a cloud to one quantum, an input and not derived (Section 5.3), and fixes no Γ , so (1), (2) and (3) are computed as functions of Γ until the line that fixes it is written.
- (5) *The lattice constants, $3G(0) = 0.7582$, $3/(4\pi)$* far away and light's $1/\sqrt{3}$ Links per interval, the six Ports' own, exact and not nature's, entering every derivation above (COMPUTED).

8.3 The hypotheses under their own names

Every hypothesis stands outside the law under its own name until it passes the three tests or the owner admits it; each says what would decide it [1, the hypotheses under their own names].

Hypothesis 1 (the conversion, the weak force). A click that takes a quantum of one family and gives it as quanta of others, the rotations adding, is the whole weak interaction, one Node's act, no field and no massive vector, and no family's row declares anything for it.

Hypothesis 2 (Gamma is not constant). $\Gamma_{t+1} = \Gamma_t + 6$, the step the six Ports of a Node and no free number; the rule's physics depends on the ratios c/Γ and m/Γ alone, so every pair steps with Γ and the wall and the amplitude unit are derived from Γ_t each interval; a pixel of a fixed count falls under the rising edge and dissolves, the distinct bound counts grow with Γ , light does not drift while the clocks of pixels do, the rule's Hubble rate $H = (6/\Gamma) |d \ln \omega_b / d \ln (c/\Gamma)|$, and $\Gamma = 6t$ reads the universe's age in intervals. The time road's price: a stepping Γ is Dirac's road, G falling as $1/\text{age}$; nature bounds G 's drift today under 10^{-13} per year against 7×10^{-11} per year for a G proportional to $1/\text{age}$, and in the law a coupling that decays weakens the bodies' binding with it unless the binding is a row of its own, which the two rows made it [1, Gamma is not constant].

Hypothesis 3 (the three distances are the rule's). That α , m_e/m_P and m_p/m_e are derived; the algebra is closed on one root, two named assumptions, one assumption of the universe and one hypothesis, with no number but gravity's E_g , and is not closed toward nature by name [1, the tree].

Hypothesis 4 (the hill holds the body). A body stands as a stable extended fixed point where a held row read with the opposite sign, of reach shorter than the hollow's, is held beside the hollow: below the hill's reach both are Coulomb-like and the hill forbids the collapse, between the two reaches the hollow binds, and the body's size lies between the reaches. Decided by a run of the generator with the third row.

Hypothesis 5 (the cooling). A body above its minimum sheds its excess as quanta of its own family, messages of matter moved by the count's line, and radiates light by its write while it breathes, the shedding stopping at the stable body theorem's minimum for its count; the engine sheds by the count's line alone, and whether a body settles at the minimum is a run's finding. The number of quanta a cloud gives to settle: for a Gaussian cloud twice too wide in its own massless well, 13 percent leave on a chain, 20 on a plane, 23 in a box (DERIVED, a blind expectation) [1, the number of quanta a cloud gives to settle].

Hypothesis 6 (the anti-body is the count's sign reversed). The anti-body of any body is the same record with its count's sign reversed, a body of holes, which the count's line makes and conserves; a neutron and an antineutron are both real records and differ by the sign of their count, and the baryon number is the count's sign. Decided by a lay of a body of holes and its clicks against nature's antineutron, distinct from the neutron.

Hypothesis 7 (the second level's amplitude is a count of its own). The second level's amplitude quantised as a count of its own, a circular quantum the unit and a body's charge its number of

circular quanta, so that $|q_p| = |q_e|$ would be derived where the law as written leaves a body's charge continuous (Section 3.4); the count's line as written does not do it and Theorem 2 may not allow it.

Hypothesis 8 (the board grows on demand). One Node starts and Nodes are added on demand only: the unbounded board with 0 beyond the front, since Rule3 needs nothing else and a level leaves 0 only when a neighbour was not 0 the interval before. Computed by the advisor on one octant with the far faces never reached: a static source switched on builds its rest behind the front with no ringing on the rows without a gap or with a small one (the massless row at the seed 758 at eighty intervals, its rest 758 exact; the binding holder 746, its rest 746), Huygens on the lattice; the polarisation row rings and radiates it away within a hundred intervals, its troughs sizing c_{vac} ; and a seed of one Node is not a start the law accepts on any board, so the first act of a world stays the generator's lay. What it buys: no faces and no mirrors, no sink question, the massless rows' rest built by the run itself, and the universe's size its age in Links, growth and not stretching, from which Hubble's redshift does not come. Not decided; the owner's word [1, the board grows on demand].

Beside them, by name: the third way of gravity (the massless row's level as a count of resting clicks relaxed each interval, which trades the exact inverse or the causality; Le Sage's shadow refused, the drag on a moving body and the absorbers' growth), a controlled phase of order one between two records (which Theorem 2 excludes from the acts alone), and a cross-axis read or a current's read (frame dragging, the induction), which Rule3's line reads nowhere [1, the fork, decided; what is open, item 15].

8.4 What is open, and what is not claimed

The law's document keeps its open items by name; an item the law's own words closed stays as one line saying why, a decided line stands against nature until tested, and a line decided with a reservation stands until one of its reopeners [1, what is open].

- 1 to 3. *The proton and the electron, the two branches, the fall and Q/M* , closed: the body of nature is the cloud, its three masses one, the proton a cloud of 1,836 and the electron one quantum, m_p/m_e the count's ratio fixing no Γ ; the coupling through the pace is universal and the response is the mode's own, so the equivalence principle holds for the bodies of nature and not for collapsed matter, whose response is suppressed (Sections 5.3 and 6).
4. *The lone quantum's detection*, decided and against nature until tested: the click ends nothing and absorption is the read by the sense and the write, so a lone quantum spread over many Nodes never clicks at a cold detector Node, clicks at a warm one by its remainder as the intensity's share, and clicks once at a resonant absorber, locally; what stays against nature is the two takers, with a number (Section 3.7); the two slits' dilution series on a warm board and the absorption look test it, and their result is the law's answer, not a line to add.
- 5 to 8. *The colours and the third*, closed as absent; *the universe expands*, closed as a statement about boards (a board closed on every axis has no rest and its level drifts, the finite board's word); *the unread parts*, closed as out; *the builds*, closed as decisions.
9. Γ and T : Γ the one scale of the law, as the Planck length is nature's, free and large against a proton's count; T physical through $\sqrt{T}/\Gamma$; what fixes either is open.
- 10 to 12. *Radiation reaction*, closed by the sense's build (whether a breathing body cools is a run's finding); *the mass defect*, closed as the law's reading (Section 5.2); *the neutron and the antineutron*, closed by the hole (Hypothesis 6).
- 13, 14. *The generator's branch and the start's sources*: the start sourced by the wells, the fixed point of the wells under the rest they make, is the pixel round's line; *the tensions' start*: the start

sources the time parts alone, so the tensions start at 0 and ring up to their rest over the first intervals, a transient that is no wave of the body.

15. *A cross-axis read or a current's read*, a hypothesis under its own name.
- 16 to 18. Whether a body at rest jitters when its current's swing reaches $T/2$ (on sixteen Nodes the swing over 400 intervals is one percent of $T/2$); the self-source's cubic term, not in the engine; the stable body theorem's Hessian beyond the dilation for a set of held rows and the functional beyond first order, and whether the hill holds the body and the cooling brings a body to it, a run's findings.
19. *What fixes Γ* : the mass ratio of clouds is a count's ratio and fixes no Γ ; the constants of Section 8.2 are computed as functions of it until the line that fixes it is written; that line is open.
20. *The hierarchy of gravity and electricity*, decided with a reservation (Section 4.4): the finding of every divisor 1 kept as the law's word on the road not taken; the reopeners the three tests, the bodies not standing under the screened sum, the charged cloud look reading nature's other way, and the finding of scale.
21. *The laid row's inverse*, decided: a held row laid each interval from the wells has no whole-run inverse (the paces of an interval read a level that is a function of a level no later state holds), so every holder of the file steps by Rule3 at its pair and inverts by its own direction, and the whole run goes back in time (Section 2.9).
22. *The vacuum content*, decided (Section 4.5).

The findings against nature, in one place. Written as they stand, each the law's own and none a rounding: the two takers, both rising on one quantum with the probability $f_1 f_2$ (Section 3.7); the bare detector Node in a gap that changes nothing (Section 7.4); the charge's unit not derived, a body's charge continuous (Section 3.4); the charged cloud heavier than the neutral one where nature's neutron is the heavier (Section 5.3); under every divisor 1 the electric force $1/1,836^2$ of the gravity where nature's is 10^{36} the other way, decided away by the two rows with the caveat, and the finding of scale which the two rows do not touch (Section 4.4); the light's self-focusing by its own well at the coupling 1, the advisor's reading of the two slits' centre (Section 7.4); and the bending's 1.9 against 2, the band's slope (Section 6).

Not claimed. Any agreement with nature beyond the one row shown, the two slits' fringes; that nature is so; the Lorentz group, which is no symmetry of the lattice; Bell's correlation and the two takers' anticorrelation; a controlled phase of order one between two records; the fine-structure constant, the mass ratio and what fixes Γ ; the spin $1/2$; the count of gravitational levels, which is the universe's history; a body's settling, which is the hypothesis of the cooling. Every hypothesis of the paper is named as one.

9 The engine that computes it

The engine, Universe24 [5][2], is the law's implementation and nothing else: a GameBoard of Nodes, every family's NodeState at every Node, and one interval of acts in the law's order, each a call of Rule3 on whole-board arrays in 64-bit integers, every neighbour read through a Port. It holds no number, no formula, no family's name and no flag; every physical value comes from the run's files; no act of the interval takes a square root and no act reads a Node to decide anything. It is six modules under 400 lines each on a core of Rule3, the Ports and the working bound; every family is a row of the universe file found by its name, and a new family is a row and no code [2, the engine; the NodeState; how to add a family][3, the genericity of the families; not one number in the engine].

The files. *The universe file* declares the integers (Γ , T , the width) and the families, each its name, its pair and, for a held family, what it holds and its divisor; the rule derives the rest. *The world file* declares the board's shape and its faces (periodic, open or closed, the inner faces with their gaps), the intervals, the bodies by their family, their Nodes with their counts and the quanta of other families they hold, the messages by their family, axis, wave number, amplitude and envelope, and the detectors by their Nodes or the body they name; a missing key is refused by name, and no default is written. *The mode file* beside the world holds the generator's levels of every body and message. *The expectation file* holds the blind row, every number the file's [2, the loader and the files].

A run. The generator lays the bodies and the messages; the run is headless, one output file per world with the verdict (LAWFUL, or REFUSED with the reason, the guard's at load among them), the intervals run, one `click` line per quantum with its family, its detector, the Node it entered and the Port's axis and side, and the books (a GameBoard diagnostic, the least Link pace of the final state among them); a reader of the clicks reads the rises per detector Node over a window against the expectation file. A look runs only after the back-in-time gate says MATCH on its own world over its intervals [2, the output; how to run a world].

The gates. Every pull request runs lint, formatting, strict types and the tests of the changed files and their consumers; the gates hold the rule's arithmetic to one function, every shift across a Link to the Ports, every integer literal of the engine and its tools to the law's own, no root anywhere, every module to integers alone, the acts against Rule3 by hand, the interval on a closed cube (the count conserved, every holder at an isotropic rest, the 48 symmetries in every part, the gate MATCH), the tension as Rule3's own conservation of the current, light born by the write, the rest of every pair as its line's own fixed point, the generator's bodies and its message lay against the real numbers, the documents' headings and the repository's language. A behaviour change needs a dedicated test; a physics change starts from the law's line [2, the gates].

To test a hypothesis. Write its line in the law's document first, under its own name, with the three verdicts; the advisor computes the blind number from the engine's own lines; the worker builds it as one pure function of arrays that calls Rule3 alone, with its test, on one short branch; the world and its blind expectation are written before the run; the run's reading is compared with that number, a difference being a defect of the engine and never a finding, and a number outside its band with no defect found naming the missing law as a hypothesis under its own name. No expectation is rewritten after a look, and no body's number is tuned to a result [4, the blind number first].

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Data and code availability. The law's document, the engine that computes the rule and the figures' scripts are archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [5], under the MIT licence; the one run this paper cites, the two slits, is cited through the law's document and names its commit there.

Author contributions. The author laid down every postulate and decision of the model, directed the work and wrote the paper.

The use of a large language model. The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under his direction and instructions, for the software, the computations, the mathematical derivations and their checks, the physics of the law's document and the drafting of the text; the model is not an author; the author reviewed and edited every statement and takes full responsibility for the content. No figure was made by a generative tool; every figure is drawn by a script from the definitions or from the documents' rows.

References

- [1] The law of the GameBoard, `docs/ALGEBRA.md` of the archive [5], cited by the name of its section: the GameBoard and the families (the objects, a family's declaration, the interval, readings and measurements); Rule3 (the line, the direction, the four acts, the paces, the bound, the self-source, the conserved form); the click (the booking, the count's line); the primitives; the stable body (what a body is, the well, the functional and the dilation, the surplus leaves, the generator, the velocity); what the law says of nature (the reading, the hypotheses under their own names, the postulates, the constants, the rows against nature); what is open.
- [2] The engine, `docs/ENGINE.md` of the archive [5]: the words, the NodeState, the interval, the loader and the files, the output, how to run a world, how to add a family, the gates.
- [3] The decisions, `docs/HIGHLIGHTS.md` of the archive [5]: the model owner's decisions in force, one line each, cited by the line's name.
- [4] The shared way of working, `skills/workflow.md` of the archive [5]: the three tests, the six verbs, the blind number first, the short procedure.
- [5] A. Gonen, Universe24: a discrete simulator of local physical laws (Reality Theory), Zenodo (2026), <https://doi.org/10.5281/zenodo.22738746> (the concept DOI, resolving to the latest version).